



VIBRATING MACHINERY FOUNDATIONS ON SOIL (USING FREQUENCY INDEPENDENT CRITERIA)

PURPOSE

This document establishes guidelines, recommended procedures, and sample calculations for the design of soil supported foundations for large reciprocating compressors, centrifugal compressors, and other similar vibrating equipment. The vibration analysis is based on frequency independent soil stiffness and damping in a procedure originally described by Richart and Whitman.

SCOPE

This document provides the following:

- Basic theory behind frequency independent criteria.
- A description of the formulae and sequence to perform a vibration analysis.
- A discussion of acceptable design results.
- Additional design conditions including situations not normally encountered.

A sample design is included as an aid in producing actual designs. A centrifugal compressor foundation is analyzed and designed by and confirmed with a computer program.

APPLICATION

This document applies to vibrating equipment generally weighing more than 5,000 pounds supported on rigid block foundations.

THEORY

Reciprocating compressors generate harmonic unbalance forces of substantial magnitude at low operating frequencies.* The operating frequencies often lie close to the natural frequencies of the foundation in its various modes, thus creating resonance or near resonance response conditions in the machine foundation system. The magnitude of vibration amplitude at or near resonance conditions becomes a controlling criteria. Therefore, inclusion of the effects of internal and geometrical damping during oscillation becomes a very important consideration and can only be accomplished by using the elastic half-space theory. In this theory, the footing is assumed to be rigid, to rest on the surface of an elastic half-space, and to have simple geometrical areas of contact, either circular or rectangular. The half-space itself is assumed to be a homogeneous, isotropic, elastic semi-infinite body which is often called simply the half-space. This theory includes the dissipation of energy throughout the half-space by geometrical (radiation) damping. This loss of energy occurs through transmission of elastic wave energy from the footing to infinity. The method is an analytical procedure which provides a rational means of evaluating the spring and damping constants of the soil/foundation system for incorporating into a lumped-parameter, mass-spring-dashpot vibrating system.

This method is the current state-of-the-art for the dynamic analysis of footings resting on soils.



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- * Reciprocating compressors in theory generate imbalance forces at the machine operating frequency (denoted as primary), twice this frequency (denoted as secondary or first harmonic) and all higher harmonics. Only the primary and secondary unbalance forces/moments are ever considered in design. However, the secondary unbalance forces/moments can be substantially greater than the primary effects in many cases.

Knowing the mass, spring constant and damping ratio for each of the 6 dynamic degrees of freedom (modes) of the system**, the response of the system in each mode can be readily determined for any given forcing function. Although relatively simple to analyze, the major problem has always been the determination of reasonable values for the 3 dynamic parameters; mass, spring constant, and damping ratio. The latter 2 parameters are functions of 3 soil properties, shear modulus, Poisson's ratio, and mass density.

- ** The 6 degrees of freedom are defined as translation along 2 horizontal axes (parallel and perpendicular to the shaft centerline) vertical translation, rocking about the 2 horizontal orthogonal axes and torsion about a vertical axis.

The development of the theory dates back to a landmark paper by H. Lamb in 1904. He first studied the response of the elastic half-space as it was excited by oscillating vertical forces acting along a line. Thus, he established the solution for 2-dimensional wave propagation. He extended the solution to horizontal forces and 3 dimensions for a point loading. The oscillating vertical force at the surface has been termed the dynamic Boussinesq loading. By integration of the force over a finite area, the dynamic response of a footing can be described. This integration was carried out by the German engineer, E. Reissner, in the early 1930s, in his efforts to provide a basis for evaluating the dynamic response of a vibrating footing (as measured in the laboratory by a mechanical oscillator) as it was influenced by properties of the soil. He chose to use Lamb's elastic half-space to represent the soil in which shear modulus, Poisson's ratio, and mass density are sufficient to describe its elastic parameters. He concentrated on a circular footing with uniform pressure distribution and obtained an analytical solution for periodic vertical displacement of the loaded surface.

Subsequent researchers, P. M. Quinlan, T. Y. Sung, and T. K. Hsieh, extended the work to consider changes in pressure distribution over the footing contact area and improvements in the expressions for geometrical damping. Hsieh's most important contribution, however, was his representation of the dynamic response in the form of the dynamic equation of motion of a single degree of freedom lumped mass oscillator with the special condition that the damping and stiffness terms are frequency dependent.

Subsequent investigators, J. Lysmer, J. R. Hall, and others, developed analogs and certain simplifications which resulted in expressions for constant damping and stiffness. This was accomplished through use of modified dimensionless mass ratios and other empirical adjustments which furnished the bridge between the half-space theory and mass-spring-dashpot system. It is not surprising, then, that the formulas for damping and stiffness contain non-integer factors. It is the result of a best fit match of dynamic response between the half-space theory and the lumped-parameter analog. Although not apparent, the appearance of the dimensionless mass ratio term in the formulas is in fact a frequency dependent factor. It is for this reason that we retain a special non-constant empirical



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correction for rocking response; all other modes of motion are represented by a constant correction term.

Cylindrical foundations embedded into an elastic half-space have been evaluated by H. Tajimi's elastic theory, by J. Lysmer's and R. Kuhlemeyer's finite element method, by K. H. Stokoe and F. E. Richart, Jr. model tests, and by an approximate method by M. Novak, Y. O. Meredugo, and M. Novak. Embedment increases both stiffness and damping; although, additional work remains to be done on the effect on torsional vibrations. The results of this work were summarized by R. V. Whitman, and is the source of the embedment correction factors for modal spring constants and damping ratios given herein. The benefits of embedment can only be achieved, however, if the foundation adheres firmly to the soil at the base and at the sides.

Extensive field tests on model and prototype footings have shown very good agreement with the vibration response predicted by theory. Nevertheless, be reminded that soil mechanics and foundation design is an imprecise science. Soils are variable. They are not homogeneous. Dynamic soil properties can vary in short distances as much as 25 percent from those reported by carefully conducted tests. Hence, it is incumbent upon the analyst to consider the potential effect of possible variation in soil parameters from those reported. The design should be investigated and shown acceptable for a reasonable variation of parameters. This step is especially important for foundations supporting vibratory equipment because in this unique situation, it is possible to measure the vibration response of the foundation during machine operation and, thereby, verify the adequacy of the foundation design. Such vibration tests are common and readily conducted with portable hand held instruments. Thus, our obligation is to use the best available theory, procedures, and data available (as outlined herein) to achieve an acceptable design within the context of the tools and information at our disposal.

DEFINITION OF LUMPED PARAMETERS

Mass

Clearly, the mass and mass moments of inertia of the equivalent lumped parameter system should include the mass of the foundation and supported machinery. In addition, there is a school of thought that proposes a certain in-phase vibrating soil mass should also be included. R. V. Whitman, F. E. Richart, J. R. Hall, and R. D. Woods demonstrate that such in-phase soil mass is quite inappropriate for the half-space theory as developed for the lumped parameter system and, they argue, is either negligible or rendered unimportant by high damping near resonance. Accordingly, the lumped mass chosen for the analysis is limited to the mass of the foundation, soil above and supported on the footing (if any), and the supported machinery.

Spring Constant

The spring constant is usually the most important parameter of the 3 parameters utilized in the lumped parameter system. The value of the spring constant affects the frequency,



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magnitude of motion at resonance, and the magnitude of motions occurring at frequencies well below resonance. Attachment 01 gives formulas for spring constants for the various modes of vibration of circular and rectangular footings resting on the surface of an elastic half-space, that is, on the surface of the soil.

Note: Attachment 02 is used in conjunction with the expressions for rectangular footings in Attachment 01.

Foundations with embedment in the soil exhibit increased stiffness and for these cases, the spring constants are modified by multiplying the expressions from Attachment 02 by the coefficients N of Attachment 03, Table 2.

For the great majority of cases, one can evaluate spring constants and damping ratios (see following Section) for rectangular footings by considering an equivalent circular footing. This approach can simplify the calculations since the expressions for damping ratio are given only for circular footings. Attachment 03, Table 2, gives equations for calculating an equivalent circular footing radius, r_o , for a corresponding rectangular footing of dimensions b by L .

Thus, if the values of r_o are substituted into the spring constant equations for a circular base, one obtains essentially the same results as using the equations directly for the rectangular base. However, this equivalence holds true only for certain limits of the plan aspect ratio of a rectangular foundation in 3 modes of vibration as given in the following: For vertical vibration, spring constants are comparable for:

$$0.33 \leq \frac{L}{b} \leq 3$$

For horizontal vibration, spring constants are comparable for:

$$0.17 \leq \frac{L}{b} \leq 6$$

For rocking vibration, spring constants are comparable for:

$$0.33 \leq \frac{L}{b} \leq 3$$

No such restrictions apply to the torsional mode of vibration.

If the rectangular foundation shape meets these requirements, one can first calculate the equivalent r_o values from Attachment 03, Table 2, and use these to obtain spring constants from Attachment 01 and damping ratios from Attachment 04, Table 1. Otherwise, one must use Attachment 01 and Attachment 02 for vertical, horizontal, and rocking spring constants; use Attachment 03, Table 2, in conjunction with Attachment 01 for the torsional spring constant; and, finally, use Attachment 03, Table 2, in conjunction with Attachment 04, Table 1, for calculating damping ratios.



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Damping

The dashpot in the single degree of freedom system represents the damping of the soil in the foundation soil system. There are 2 types of damping in the real system:

- Radiation damping (occasionally called geometrical damping)
- Internal damping

The latter is of lesser importance in most cases.

Radiation damping involves the loss of energy through propagation of elastic waves away from the immediate vicinity of the footing. Expressions for the damping ratio, D , have been obtained for rigid circular foundations resting on the elastic half-space. Using the equations given previously in Attachment 03, Table 2, the rectangular base of dimensions b by L can be converted into an equivalent circular base of radius r_o and the expressions in Attachment 04, Table 1, can be used to determine the damping ratio, D , for foundations without embedment.

Small rocking mass ratios, $B_R < 5$, require a correction on rocking inertia, I_R and B_R , used in the damping calculation. If $B_R \geq 5$, omit the correction. If $B_R < 5$, the procedure is as follows:

- Calculate B_R using the expression in Attachment 04, Table 1.
- Determine the correction factor, n_R , from:

$$n_R = \frac{1.219}{B_R^{0.169}} \text{ for } B_R \leq 1.0 \quad n_R = \frac{1.219}{B_R^{0.0758}} \text{ for } 1.0 < B_R < 5$$

These equations are developed from data given in Vibration of Soils and Foundations.

- Calculate effective B_R as:

$$B_R(\text{effective}) = n_R B_R(\text{calculated from Attachment 04, Table 1})$$

Note: Use I_R , not $I_R(\text{effective})$, to calculate B_R .

- Use $B_R(\text{effective})$ to determine effective damping D_R from Attachment 04, Table 1.
- Calculate effective I_R as:

$$I_R(\text{effective}) = n_R I_R(\text{calculated})$$

- Use $I_R(\text{effective})$ in place of $I_R(\text{calculated})$ in all subsequent calculations.



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Foundations with embedment also exhibit increased radiation damping. Damping ratios for embedded foundations can be determined by multiplying the radiation damping ratios of Attachment 04, Table 1, by the coefficients S in Attachment 04, Table 2.

Internal damping involves energy loss during stress reversals. For dry or relatively dry cohesionless soils, the energy loss is caused by sliding between soil particles. With wet or saturated soils, energy loss occurs as a result of relative motion between the soil skeleton and pore fluid. Available information indicates that a reasonable value for the internal damping ratio, D , is 0.05 for all conditions.

A comparison of radiation and internal damping indicates that radiation damping is much higher than internal damping for vertical and horizontal vibration. For rocking and torsional vibration; however, radiation damping is usually low and may be of an order close to internal damping. In all instances, internal damping ($D = 0.05$) should be added to radiation damping.

For foundations on soil, the total of calculated radiation plus internal damping should be used in the analysis, but the sum will not exceed the following values:

Mode	Maximum Total Damping For Analysis
Vertical	0.95
Translation	0.60
Rocking	0.40
Torsion	0.20

Further to the above, in order to realize the theoretical increases in stiffness and damping arising from embedment as given by the foregoing expressions, it is essential that there be intimate contact between the sides of the foundation and adjacent compact soils. This can be ensured by pouring the foundation directly against undisturbed soil or, if the sides are formed, requiring the backfill to be compacted to 95 percent of Modified Proctor (cohesive soils) or 85 percent relative density (cohesionless soils).

Embedment in cohesionless soils ensures continued firm contact between the soil and the sides of the foundation. Vibrational energy transmitted to such soils continuously provides compactive effort to maintain the effectiveness of the embedment. Expansive, cohesive soils on the other hand can shrink away from the sides of the foundation and, thereby, reduce or even eliminate all benefits of embedment. Shrinkage on the order of less than 0.001 inch (25 microns) is sufficient to negate embedment benefits. Therefore, foundations embedded in cohesive soils must be analyzed and shown acceptable for the 2 extreme conditions of:

- Full embedment contact
- No embedment benefit



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Foundations Overlying an Elastic Layer

If the soils underlying the foundation are layered and have a hard stratum of soil or rock at a shallow depth below the footing, the formulas for spring constants and geometric damping (based upon a semi-infinite elastic half-space) may be in error. The radiation of energy from the footing is impeded by the presence of the rigid layer and part of this elastic-wave energy is reflected back to the footings, not radiated away as assumed. Theoretical and model studies of the vertical, horizontal, and rocking oscillations of circular footings indicate an increase in the spring constant and a decrease in damping as the layer thickness ratio, T/r_o , decreases, where T is the thickness of the elastic layer overlying the (infinitely rigid) bedrock. As the elastic layer becomes thinner, the vertical spring constant, K_{v1} , increases approximately as:

$$K_{v1} \approx K_v (1 + r_o / T)$$

The reduction in damping ratio, D_{v1} , is given in Attachment 05.

Further, the horizontal and rocking spring constants are increased approximately as:

$$K_{H1} \approx K_H (1 + r_o / 2T) \text{ and} \\ H_{R1} \approx K_R (1 + r_o / 6T)$$

as the elastic layer becomes thinner. Because these increases in spring constants are not as strongly influenced by the r_o/T factor as is K_{v1} , it follows that the damping ratio decreases would be smaller for these modes than shown in Attachment 05. It is recommended that damping ratios for horizontal and rocking modes be taken from Attachment 05. Values in Attachment 05 were derived by adjusting the values from Attachment 05 to account for the relative change in spring constants for vertical, horizontal, and rocking modes as the layer thickness ratio changes.

The damping limits given in the Damping Section should also be reduced by the factors given in Attachment 05. Information is not currently available for the influence of layer thickness on torsional stiffness and damping. It can be assumed that torsional stiffness and damping are influenced in the same manner as the horizontal stiffness and damping.

Modifications to spring constant and damping ratio given in this section apply only to foundations without embedment. "Vertical Vibration Of Embedded Footings" and "Dynamic Stiffness Of Circular Foundations" contains procedures to treat embedded foundations overlying an elastic layer and to evaluate modification to damping for layer thickness ratios other than those given in Attachment 05.



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INFORMATION NECESSARY FOR DESIGN

Soil Properties

An examination of the expressions for the spring constant and damping ratio for the various modes of vibration reveals that 3 soil properties are required for the analysis:

- Soil mass density (ρ)
- Poisson's ratio (μ)
- Dynamic shear modulus (G)

It is of paramount importance to have quality data on these properties for the dynamic analysis. These properties must always be requested from the geotechnical consultant. We must insist on low strain (shear strain $\leq 10^{-5}$), down hole seismic survey* to be conducted by qualified geotechnical personnel with proper equipment.

There will be situations when shear wave velocity tests, of whatever kind, have been previously conducted by the client's consultant. The reliability and accuracy of such data for use in design must be carefully evaluated on a case by case basis. Refer to Specification 000.210.CVS02010: Geotechnical Investigation, for additional clarification.

Such tests should be made to a depth equal to the largest equivalent circular foundation radius, r_o , given in Attachment 03, Table 2. If r_o is not known, the tests should extend to a depth of 30 feet (10m). If possible, the tests should be made at the actual foundation location.* The results of the seismic survey provide the profile of shear wave propagation velocity, V_s , under the foundation which in turn will establish the profile of dynamic soil shear modulus, G .

Note: $G = \rho V_s^2$, where ρ = soil mass density = γ / g

In some cases, foundation locations may not be known before the plot plan is developed. Judgment should be exercised in such cases and test locations determined in consultation with the geotechnical consultant. It is not intended to require a second field mobilization to perform the seismic survey after the plot is finalized.

For design purposes, it is necessary to calculate 4 effective G values (translation, rocking about 2 axes and torsion) for use in the appropriate stiffness formulas, Attachment 02. These effective G should be obtained for each mode by weighing the G profile for layer thickness and with a triangular distribution varying from zero weight at the underside of the foundation. Refer to Attachment 06.

Both compression (P) and shear (S) wave propagation velocities can be measured in a down-hole seismic survey. Their ratio can be used to calculate Poisson's ratio. However, such calculations involve small differences of rather large numbers, and significant errors are possible. This approach is further complicated because of several other factors, the most important of which is that the P wave velocity in the soil cannot be measured below



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the water table level. Below the water table, the P wave is transmitted through the fluid (not the soil) at high speed (5,000 ft/sec) (1,500 m/sec) and, thereby, masks the true soil transmission velocity. The shear wave speed through soil, on the other hand, is not influenced by the presence of water. Because of these considerations, one should be strongly influenced by and rely upon the typical values for Poisson's ratio shown in Attachment 07 unless there are compelling reasons to use values other than these. Generally, Poisson's ratio is seen to vary between 0.25 to 0.35 for cohesionless soils and between 0.35 to 0.45 for cohesive soils. Consequently, for design purposes, little error is introduced if Poisson's ratio is assumed as 0.33 for cohesionless soils and 0.40 for cohesive soils.

Keep in mind that near surface tests can be influenced by nearby foundations, paving or buried organic material which can lead to erroneous measurements of wave speed, both high and low. Further, the soil immediately above the water table may be subject to capillary tension which effectively increases overburden pressure leading to higher measured wave propagation velocity and correspondingly higher shear modulus. Accordingly, when examining the profile of shear wave velocities, one should neglect apparent localized increases at or near the water table level or near the surface. In general, the profile should reflect a smooth, rather gradual increase in shear wave velocity with increased depth, except where rock or other firm stratum is encountered. Typical values of soil shear modulus are given in Attachment 07 as a guide.

Machine Data

The machine data required for foundation design are as follows:

- Dimensions of machine base.
- Location, type, and size of anchor bolts and support plates.
- Weight and location (horizontal and vertical) of CG (Center of Gravity) of combined machine assembly and each component.
- Mass moments of inertia of machine (about 3 axes).
- For reciprocating machine, primary and secondary unbalanced forces, including their location.
 - Vertical (translation, Q_o)
 - Horizontal (translation, Q_o)
 - Vertical couple (rocking, M_o)
 - Horizontal couple (torsion, T_o)
- For centrifugal machine, dynamic imbalance force and location for each rotor. In lieu of such supplier data, the machine unbalance force, Q_o , can be determined from:

$$Q_o \text{ (kips)} = \frac{\text{Rotor Speed (rpm)}}{6000} \times \text{Rotor Weight (kips)}$$

And will be applied transverse to the rotor shaft midway between bearings. In the latter case, the rotor weight is required from the supplier.



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- Machine operating speed or speed range.
- Supplier foundation recommendations (if any).

DESIGN APPROACH

Foundation Geometry

Generally, the compressor foundation consists of a concrete pier and mat footing. Centrifugal equipment, however, does not always require a mat. Minimum pier dimensions and minimum mat thickness are often furnished by the compressor manufacturer. When this information is lacking, the pier should extend at least 4 inches (100 mm) beyond the compressor base or satisfy the edge distances required for anchor bolts as specified in Design Guide 2.2. Whenever possible, anchor bolts for large compressors should be 12 inch (300 mm) minimum from bolt centerline to face of concrete.

The width of the mat, dimension perpendicular to the crankshaft, should be at least 1- ½ times (use ¾ for centrifugal machines) the distance from the centerline of the shaft to the bottom of the foundation. The length of the mat or dimension parallel to the crankshaft should be at least 2 feet greater than the length of the pier.

The horizontal eccentricity, in each principal horizontal direction, between the center of mass of the machine foundation system and the centroid of the foundation contact area should not exceed 5 percent of the corresponding foundation dimension. This limitation ensures essentially uniform static settlement and avoids several coupled response modes.

As a general principle, one seeks to develop a foundation such that the CG of the combined machine foundation system is as close as possible to the lines of action of the unbalance forces.

The mat thickness must be great enough to ensure a rigid foundation. The minimum thickness of the foundation mat will not be less than 2 feet nor less than the value given by:

$$t = 0.02 \left[\frac{GL^4}{b} \right]^{1/3}$$

where

t = Mat thickness (ft)

G = Soil shear modulus (psi)

L = Cantilever length of mat - maximum distance in the direction of rocking motion from face of pier to edge of mat (ft)

b = Width of mat section undergoing bending due to rocking of the foundation (ft)



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or, in terms of metric units, by:

$$t = 0.01 \left[\frac{GL^4}{b} \right]^{1/3}$$

where

t = Mat thickness (m)
 G = Soil shear modulus (kN/m^2)
 L = Cantilever length of mat (m)
 b = Width of mat (m)

The above expression is derived from beam on elastic foundation theory using a flexural deflection pattern that approximates essentially rigid mat behavior.

For the case of several machines on a common mat, this requirement can be met by considering the mat dimensions associated with the tributary area for one machine.

In most cases, the compressor and driver are supported on one common pier. If it becomes necessary to have separate piers, it may also be necessary to increase the mat thickness to ensure adequate rigidity between the piers.

If 2 or more reciprocating machines will be installed a short distance apart, such that the individual mats would be only a few meters apart, the compressor piers must be supported on a common mat. However, a common mat supporting reciprocating compressors should not be extended to support the building columns. If conditions appear to preclude use of a common mat, discuss the matter with the Technical Manager of Structural Engineering before abandoning a common mat concept.

The low soil bearing resulting from the use of a large mat allows the mat bottom to be raised close to the ground surface, decreasing the quantity of structural materials and excavation. The top surface of the mat is sloped from the pier to the edge of the mat with a single pour. The surface beneath the high pressure piping support piers and other piers is left rough to provide aggregate interlock for resisting shear.

Reinforce all exposed faces of the compressor piers with a minimum of #5 bars at 12 inch (16 mm ϕ at 300 mm) both vertically and horizontally. The mat reinforcing is obtained by computations.

Soil Pressure

Foundations for rotating equipment are sized primarily for the purpose of minimizing vibration amplitudes. Since these foundations tend to have plan dimensions larger than required for supporting the gravity loading, soil bearing values are usually small. However, as a rule of thumb for design, the soil bearing should not exceed 50 percent of the allowable static soil bearing to obtain a footing which will economically satisfy the vibration analysis.



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Procedure for Dynamic Analysis

Once a trial foundation size is determined, the second phase of design is the dynamic analysis. Following are the steps required for the analysis:

1. Calculate the lumped parameters of mass (m), damping ratio (D), and spring constant (k) for the 6 modes of vibration using the equations given previously. For the rocking and torsional modes of vibration, replace the mass with the mass moment of inertia (I) of machine and foundation about the appropriate axis of rotation passing through the CG of the combined machine foundation system.
2. Horizontal translation and rocking are strongly coupled. Refer to Attachment 08. This coupling exists for translation rocking motions in both horizontal directions. Hence, 2 coupled mode analyses are required to evaluate translation rocking response in the 2 principal directions. Torsion and vertical translations are, however, only weakly coupled and can be realistically treated as uncoupled (independent) from the other modes of motion.
3. For translation-rocking motions in the X-Y plane, calculate the frequencies of the 2 coupled modes of motion, ω_1 and ω_2 (radian/sec), as roots of the quadratic equation:

$$\omega^4 - \left[\frac{K_H}{m} + \frac{K_R + K_H Y_h^2}{I} \right] \omega^2 + \frac{K_H K_R}{ml} = 0$$

where

I = Mass moment of inertia of combined machine foundation system about the CG

m = Mass of machine and foundation system

y_h = Vertical distance between CG and underside of the foundation. For embedment, refer to note on Attachment 08.

And

I , K_R , and K_H are chosen appropriate for the direction of motion under consideration.

4. Calculate modal coordinates, u_i , given by

$$u_i = \frac{K_H Y_h}{K_H - M \omega_i^2}; \quad i = 1, 2$$

Generally, u_1 and u_2 will differ in numerical value and in sign.

5. Calculate generalized masses given by:

$$\overline{M}_i = m u_i^2 + I; \quad i = 1, 2$$



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6. Calculate generalized forces for the primary unbalanced forces and moments given by:

$$\overline{F}_i = Q_0 U_i + M_0 + Q_0 Y_e; \quad i=1,2$$

where

y_e = Vertical distance between the CG and the machine shaft centerline and
 Q_0 and M_0 correspond to the direction of motion under consideration.

7. Calculate the equivalent modal damping ratios, D_i^* , given by:

$$D_i^* = \frac{u_i^2 K_H D_H + K_R D_R}{u_i^2 K_H + K_R}$$

8. Calculate the maximum steady state response values, q_i , of the generalized coordinates for the primary unbalanced forces/moments given by:

$$q_i = \frac{F_i}{M_i} \left[(\omega_i^2 - \Omega^2)^2 + (2D_i^* \omega_i \Omega)^2 \right]^{-1/2}; \quad i = 1, 2$$

where

$$\Omega = 2\pi f$$

f = Primary frequency of excitation

9. Calculate phase angles, α_i , given by:

$$\alpha_i = \tan^{-1} \left[\frac{2D_i^* \omega_i \Omega}{\omega_i^2 - \Omega^2} \right]; \quad i = 1, 2$$

10. Using the phase angles computed in step 9, determine U_0 and Θ_0 for the direction of motion under consideration by vector addition of the modal components, $U_i q_i$ expressed by:

$$u_0 = u_1 q_1 + u_2 q_2$$

$$\Theta_0 = q_1 + q_2$$

With consistent units used throughout, u_0 will be expressed in the chosen length units and Θ_0 in radian.

The above vector addition can be expressed algebraically as:



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$$u_0 = (q_1 u_1 \sin \alpha_1 + u_2 q_2 \sin \alpha_2)^2 + (q_1 u_1 \cos \alpha_1 + u_2 q_2 \cos \alpha_2)^2$$

$$\theta_0 = (q_1 \sin \alpha_1 + q_2 \sin \alpha_2)^2 + (q_1 \cos \alpha_1 + u_2 \cos \alpha_2)^2$$

If the phase angles, differ by no more than about 30°, the vector addition can be α_i , replaced by a simple algebraic sum with sufficient accuracy.

11. Calculate the steady state vibration amplitudes, A, corresponding to the primary unbalanced forces and moments at points of concern using the expression:

$$A = u_0 + \theta_0 d'$$

where

d' is the vertical distance between the CG and the point where the amplitude is to be calculated (+ above the CG, - below the CG). For example, the horizontal amplitude at the compressor shaft (Refer to Attachment 08), would be:

$$A (\text{Shaft}) = u_0 + \theta_0 y_e$$

12. For reciprocating machines, steps 6, 8, 9, 10, and 11 are now repeated using the secondary unbalanced forces and moments and the secondary frequency of excitation.
13. Repeat steps 3 through 12 to determine the coupled translation rocking response in the Y-Z plane.
14. Calculate the 2 undamped natural frequencies (f_f) of the system for vertical and torsional motions using:

$$f_f = \frac{1}{2\pi} \left[\frac{kV}{m} \right]^{0.5} \dots \dots \dots \text{Vertical mode}$$

$$f_f = \frac{1}{2\pi} \left[\frac{kT}{I} \right]^{0.5} \dots \dots \dots \text{Torsional mode}$$

15. Calculate the dynamic magnification factors (M) corresponding to the vertical and torsion modes of vibration using the expression:

$$M = \frac{1}{\left[\left[1 - \left(\frac{f}{f_f} \right)^2 \right]^2 + \left(2D \frac{f}{f_f} \right)^2 \right]^{0.5}}$$



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where

f is the frequency of excitation.

Note: For reciprocating compressors, one must calculate dynamic magnification factors for both primary and secondary unbalance forces. For centrifugal equipment, only primary imbalance effects are of concern.

16. Calculate the static vertical deflection (s) and static torsion rotation (ϕ) caused by the unbalanced forces (Q_o) and unbalanced torque (T_o) using the following expressions:

$$s = \frac{Q_o}{K_v} \dots\dots\dots \text{Vertical translation}$$

$$\phi = \frac{Q_{oh} + T_o}{K_T} \dots\dots\dots \text{Torsion mode}$$

where

h is the horizontal distance from Q_o to the axis of torsion.

For centrifugal machines, the term T_o does not apply.

17. Calculate the various amplitudes of vibration, A , for the primary and secondary excitations in the vertical and torsional modes for the steady state operating condition using the expression:

$$A = s(M) \dots\dots\dots \text{Vertical translation}$$

$$A = \phi(M)h' \dots\dots\dots \text{Torsion}$$

where

h' is the horizontal distance from the axis of torsion to any point where the deflection is desired.

18. Using the above expressions from steps 11 and 17, calculate the vibration amplitudes in the x , y , and z directions at the compressor and driver bearing locations corresponding to the machine shaft(s) centerline. The contribution to the displacements from the vertical mode, the torsional mode, the coupled horizontal-rocking mode in the x - y plane, and the coupled horizontal rocking mode in the y - z plane will be added.

Further, the amplitudes in each mode and coupled modes caused by primary and secondary imbalance forces will be directly added. Hence, the total vibration amplitude at any point in the x , y , and z directions is the sum of the corresponding x ,



VIBRATING MACHINERY FOUNDATIONS ON SOIL (USING FREQUENCY INDEPENDENT CRITERIA)

y, and z contributions from each mode and coupled modes for the primary imbalance forces and moments plus the sum of contributions from each mode and coupled modes for the secondary imbalance forces and moments. The vector sum of these total x, y, and z amplitudes is not considered.

For reciprocating compressors with horizontal cylinder(s), the greatest vibration amplitude is usually the horizontal displacement (parallel to the cylinders) at a compressor or driver bearing arising from a combination of displacement contributions from the coupled horizontal rocking modes and torsion about a vertical axis.

For centrifugal machines, the above comments related to secondary imbalance effects do not apply. However, the application of dynamic forces and vibration amplitude calculations for centrifugal machines will account for the following:

- Dynamic forces from rotors operating at the same speed will be considered acting simultaneously, both in phase as well as 180 degree out of phase, to produce either maximum translational or torsional effects.
- Dynamic forces from rotors operating at different speeds will be considered independent of each other since their dynamic forces are out of phase and may only cause an occasional beat and not resonance.
- Amplitudes of forced vibration also will be determined at manufacturer specified minimum and maximum operating speeds.
- For variable speed machines, different speeds within the operating range will be analyzed, each one selected so as to produce resonance with one of the vibration modes of the foundation. Each such speed is given by:

$$f = \frac{f_f}{[1 - 2D^2]^{0.5}}$$

19. For multiple machines on a common mat, the vibration amplitude calculations are to be based upon the simultaneous operation of the maximum number of machines representing the design condition. Spare and standby machines are assumed stopped. For those machines operating, all unbalance forces and moments are to be assumed acting in phase.

Multiple machines on a common mat require careful consideration of mat flexibility as it contributes to rocking amplitude. As a first approximation, one can compute this effect in the following way:

- A. Consider a single machine supported on its tributary area of mat as an isolated foundation. That is, disregard the other machines and the remainder of the mat. Evaluate the vibration response of this system and calculate the ratio, R_1 , of total coupled translation rocking displacement at the compressor shaft to translation displacement at the mat. This represents the upper bound of rocking contribution.



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- B. Evaluate the vibration response of the common mat system using the procedures outlined herein. Calculate the ratio, R_2 , of total coupled translation rocking displacement at the compressor shaft to translation displacement at the mat. This represents the lower bound of rocking contribution because the half space solutions are based upon a rigid mat assumption.
- C. An estimate of coupled translation rocking displacement at the compressor shaft for a flexible mat is calculated as the translation displacement at the common mat times the average value of R_1 and R_2 computed above.

The final evaluation of multiple machines on a common mat should be based upon a planar finite element SAP frequency domain solution using Winkler soil springs which can properly represent actual soil and mat stiffness. A SAP solution may not be necessary for a combined mat if the upper bound value determined in Step A above, combined with other mode contributions as appropriate, is within the allowable amplitude given in the Section covering Allowable Amplitude.

In any event, the displacement from the torsional mode of vibration must be added to the above translation plus rocking displacement to obtain the total vibration amplitude.

Natural Frequency

For an undamped system, if the natural frequency equals the frequency of the forcing function, the theoretical amplitude is infinity. When damping is present, the amplitude of vibration becomes finite, but it still can be very large and, therefore, unacceptable. In order to avoid this area of high amplitudes (at or near resonance), the frequency ratio f / f_f (excitation frequency versus foundation natural frequency) should preferably be outside the range of:

$$0.7 / \sqrt{1 - 2D^2} \text{ to } 1.4 / \sqrt{1 - 2D^2}$$

The maximum amplitude of motion in any mode will occur at an f / f_f ratio of:

$$1 / \sqrt{1 - 2D^2}$$

When this range cannot be practically avoided, the analysis procedures and damping limits given herein will provide a sound evaluation of expected operating vibration levels.

Allowable Amplitude

The maximum total vibration amplitude at any bearing in the x, y, and z directions, computed in the Section for Procedure for Dynamic Analysis above, will not exceed the steady state limits show in Attachment 09. Note that all vibration amplitudes calculated in accordance with Procedure for Dynamic Analysis are single amplitude values. However, most vibration measuring instruments are used to determine vibration levels in the field



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record peak to peak amplitudes (double amplitude). Accordingly, Attachment 09 is drawn for peak to peak amplitudes and, therefore, before entering the graph, the calculated single amplitude values must be doubled.

If it is required that vibration amplitudes be evaluated for physiological effects on persons, the total peak to peak steady state vibration amplitude of any point on the foundation in the x, y, and z directions (calculated in accordance with the previous steps) will be evaluated against human tolerance levels shown in Attachment 09.

If the steady state amplitude limits given in Attachment 09 cannot be achieved within a practical foundation configuration, the problem should be discussed with the client and a mutually agreed upon compromise solution established. However, obtain concurrence with Structural management before considering any deviation from the vibration amplitude acceptance limits or any other criteria herein.

Dynamic Analysis Review Procedure

The design basis and vibration analysis of all reciprocating compressor foundations will be submitted to the Technical Manager, Structural Engineering, who will oversee an independent review and approval of the analysis, methods, and supporting data. The Technical Manager will be available for consultation during the design development and should be contacted before extensive analyses or SAP runs are undertaken.

ADDITIONAL CONSIDERATIONS

Client Specifications

Pay particular attention to possible special design requirements, more restrictive than those given herein, which may be specified in client mechanical, piping, and Civil/Structural specifications. These requirements may relate to operating vibration amplitude limits, damping restrictions, natural frequency limits, foundation dimensions, and number of machines to be considered operating simultaneously.

Interface with Mechanical and Piping

Maintain close communication with Mechanical Engineering for necessary machine data from the supplier (refer to the section on Machine Data), specifications and checking of supplier drawings and documents. Also, maintain close contact with Piping for piping hold down/anchorage requirements.

The Project Structural Engineer will draw up a needs list based upon Machine Data and the concerns expressed in Client Specifications for discussion at the initial interface job conference dealing with rotating equipment. Refer to the Department Interface Procedures Manual. When reviewing the preliminary layout and/or plot location with



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Piping and Mechanical, advise the Piping Supervisor and Rotating Equipment Engineer that reciprocating compressors with an odd number of horizontal cylinders often requires large foundation plan dimensions in order to restrict vibration amplitudes to acceptable limits. Even numbers of cylinders and centrifugal machines present much less of a problem.

At the engineering coordination meeting where all technical, commercial, and supply details are resolved in selecting a supplier, it is the Structural Engineer's responsibility to perform the following:

- Identify all needed design data and the schedule for its receipt from the supplier.
- Verify that previously developed recommendations or restrictions on the foundation are followed.
- Stress to the supplier that the compressor suction bottles, coolers, and piping must be designed, supported, or otherwise braced to ensure the absence of resonance of these auxiliary components with the primary and secondary machine frequencies.

Elevated Pipe Anchors

Note that truly rigid anchor points to resist mechanical or pulsation loads are nearly impossible to provide by tall, slender piers without the addition of congested, unsightly bracing. Thus, it is inadvisable to provide elevated pipe anchors. However, if required, such pipe supports attached to the foundation pier or mat should be designed so that their natural frequencies of horizontal vibration are either less than 0.5 times the compressor primary frequency or greater than 1.5 times the compressor secondary frequency.

Low Tuned Foundations for Centrifugal Machinery

For low tuned foundations ($f > f_i$, the machine frequency, for a short duration, will be in or very near resonance with the natural frequencies of the foundation system during startup and coast down conditions (especially turbine-drive machines). The steady state vibration amplitudes at the resonant conditions, with properly reduced dynamic unbalance forces, will be below the Coast down Limits, as defined in Attachment 09, unless specified otherwise by the manufacturer. Normally, the machine will pass through the resonances quickly and the amplitudes will not build up as they would in a steady state condition. Nevertheless, it is necessary to check this condition to avoid potential machine damage.

For purposes of satisfying this requirement, it is sufficient to evaluate the vibration amplitudes for a single resonance machine speed, f^* , given by:

$$f^* = f_f / \sqrt{1 + 2D^2}$$



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where f_f and D correspond to the foundation vibration mode which was previously determined by analysis to have the greatest contribution to the maximum total steady state operating vibration amplitude at any bearing.

The analysis at machine speed f^* will follow the steps given in Procedure For Dynamic Analysis and the calculated amplitudes will include the contributions from all modes. The unbalanced forces at the machine speed f^* will be reduced from those which exist at the normal operating speed, f . The reduced unbalanced forces, Q_o^* , to be used in the startup / coastdown analysis are given by:

$$Q_o^* = \left(\frac{f^*}{f} \right)^2 Q_o$$

NOTATION

A	= Amplitude of vibration
B	= Foundation mass ratio
b	= Foundation dimension parallel to axis of rocking
d	= Vertical distance between CG and point where amplitude is to be calculated
D	= Ratio of actual damping to critical damping
d_t^y	= Equivalent damping ratio of coupled translation rocking mode
E	= Modulus of elasticity of soil
f	= Frequency of excitation (Hz)
f^*	= Resonant machine speed during startup or coast down
f_f	= Undamped natural frequency of foundation system in the vibration mode under consideration (Hz)
$\overline{F}_i$	= Generalized force
g	= Acceleration due to gravity
G	= Dynamic shear modulus of soil
h	= Perpendicular horizontal distance from $Q(o)$ to the torsion axis



VIBRATING MACHINERY FOUNDATIONS ON SOIL (USING FREQUENCY INDEPENDENT CRITERIA)

h'	= Horizontal distance from axis of torsion to point where amplitude is to be calculated
H	= Embedment depth (2/3 of actual embedment)
I	= Mass moment of inertia of foundation and machine about the axis of rotation under consideration, passing through the <i>CG</i> of the combined machine foundation system
k	= Spring constant for the vibration mode under consideration
l	= Cantilever length of mat
L	= Foundation dimension perpendicular top axis of rocking
m	= Mass of foundation and machine
M	= Dynamic magnification factor for the vibration mode under consideration
m_o	= Unbalanced vertical (rocking) couple due to vibrating machine
$\overline{M}_i$	= Generalized mass
n_R	= Rocking mass ratio and rocking inertia correction factor
N	= Embedment stiffness increase factor
q_i	= Maximum steady state vibration response of the generalized coordinate
Q_o	= Unbalanced force due to vibrating machine
Q_o^*	= Reduced machine unbalance force at resonant machine speed
r_o	= Radius of equivalent circular footing
R	= Ratio of rocking plus translation displacement at the compressor shaft to translation displacement at the mat
s	= Static deflection due to imbalance forces acting as static loads
S	= Embedment damping increase factor
t	= Mat thickness
T	= Thickness of elastic layer



VIBRATING MACHINERY FOUNDATIONS ON SOIL (USING FREQUENCY INDEPENDENT CRITERIA)

t_o	= Unbalanced horizontal (torsion) couple due to vibrating machine
u_i	= Modal coordinates (mode shape vector assuming angular motion equals unity)
u_o	= Horizontal translation displacement of foundation at CG
V_s	= Shear wave propagation velocity
V_p	= Compression wave propagation velocity
W	= Weight of machine, foundation and soil above footing
y_e	= Vertical distance between the CG and the machine shaft centerline
y_h	= Vertical distance between the CG and the underside of the foundation, except as modified for embedment. Refer to Attachment 05.
x,y,z	= Principal orthogonal coordinate system for foundation
α_i	= Phase angle between exciting force and response α_i
β	= Coefficient for determining k for rectangular foundations
γ	= Unit weight of soil
μ	= Poisson's ratio
ρ	= Mass density of soil = τ/g
ω_i	= Undamped circular frequency of coupled translation rocking mode (radian/sec)
Ω	= Circular frequency of excitation (radian/sec)
ϕ	= Static rotation due to torsion
θ_o	= Rocking rotation of foundation

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ATTACHMENTS**Attachment 01:**

Spring Constants For Rigid Base Resting On Elastic Half-Space

Attachment 02:

β Coefficients For Rectangular Footings

Attachment 03:

Table 1. Embedment Stiffness Coefficients

Table 2. Equivalent Radii For Rectangular Footings

Attachment 04:

Table 1. Damping Ratios For Rigid Circular Foundation Resting On An Elastic Half-Space

Table 2. Embedment Damping Factors

Attachment 05:

Vertical Damping Coefficients For A Rigid Circular Foundation Resting On An Elastic Layer Of Thickness T Over A Rigid Base

Attachment 06:

Calculation Of Effective G Value

Attachment 07:

Typical Values For Soil Properties

Attachment 08:

Coupled Translation-Rocking Motion

Attachment 09:

Vibration Limits

Attachment 10:

Sample Design 1: Centrifugal Compressor Foundation Steady State Vibration Amplitudes and Coast Down Analysis



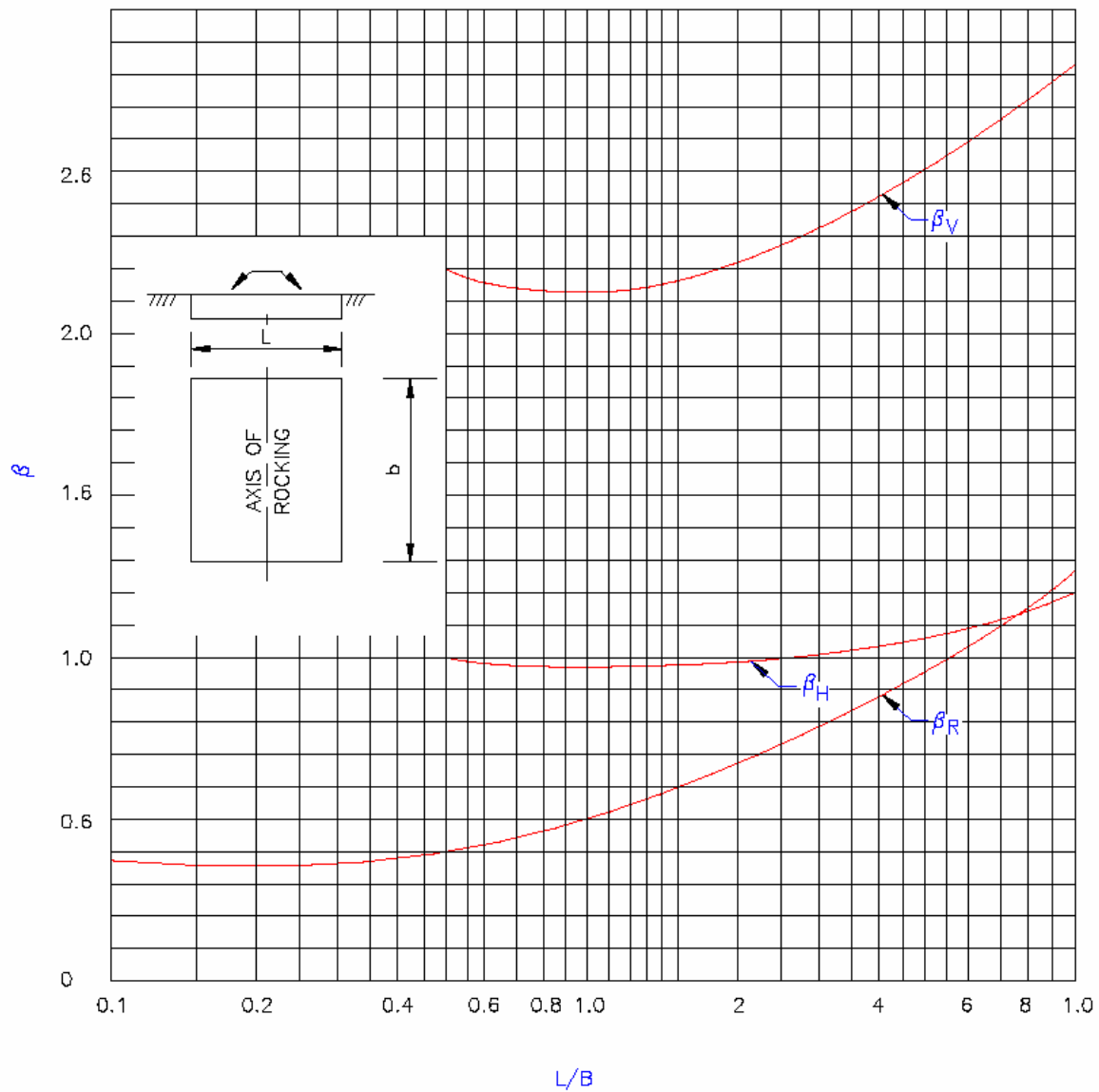
SPRING CONSTANTS FOR RIGID BASE RESTING ON ELASTIC HALF-SPACE

Motion	Spring Constant
A) Circular Base	
Vertical	$k_v = \frac{4 Gr_o}{1 - \mu}$
Horizontal	$k_H = \frac{32(1 - \mu) Gr_o}{7 - 8\mu}$
Rocking	$k_R = \frac{8 Gr_o^3}{3(1 - \mu)}$
Torsion	$k_T = \frac{16}{3} Gr_o^3$
(B) Rectangular Base	
Vertical	$k_v = \frac{G}{1 - \mu} \beta_v \sqrt{bL}$
Horizontal	$k_H = 2(1 + \mu) G \beta_v \sqrt{bL}$
Rocking	$k_R = \frac{G}{1 - \mu} \beta_R bL^2$
Torsion	Not available - Use Attachment 03, Table 2, in conjunction with (A) above.

Note!!! Dimension L is always perpendicular to the axis of rocking for formulas in Attachments 01 and 03, Table 2, and for use in Attachment 01. Values of μ are given in this attachment.



β COEFFICIENTS FOR RECTANGULAR FOOTINGS





VIBRATING MACHINERY FOUNDATIONS ON SOIL (USING FREQUENCY INDEPENDENT CRITERIA)
Table 1. Embedment Stiffness Coefficients

Motion	Spring Constant
Vertical	$N_v = 1 + 0.6(1 - \mu)(H/r_o)$
Horizontal	$N_H = 1 + 0.55(2 - \mu)(H/r_o)$
Rocking	$n_R = 1 + 1.2(1 - \mu)(H/r_o) + 0.2(2 - \mu)(H/r_o)^3$
Torsion	<i>None available; assume $N_T = 1.0$</i>
(B) Rectangular Base	
H	= Depth of mat embedment (use 2/3 of actual, to be conservative)
r_o	= Equivalent radius
μ	= Poisson's ratio

Table 2. Equivalent Radii For Rectangular Footings

Motion	Equivalent Radius (r_o)
Translation	$r_o = \sqrt{\frac{bL}{\pi}}$
Rocking	$r_o = \left(\frac{bL^3}{3\pi}\right)^{0.25}$
Torsion	$r_o = \left(\frac{bL(b^2 + L^2)}{6\pi}\right)^{0.25}$



VIBRATING MACHINERY FOUNDATIONS ON SOIL (USING FREQUENCY INDEPENDENT CRITERIA)
Table 1. Damping Ratios for Rigid Circular Foundation Resting on an Elastic Half-Space

Motion	Dimensionless Mass Ratio	Radiation Damping Ratio
Vertical	$B_V = \frac{(1-\mu)}{4} \cdot \frac{w}{\gamma r_o^3}$	$D_V = \frac{0.425}{\sqrt{B_V}}$
Horizontal	$B_H = \frac{(7-8\mu)}{32(1-\mu)} \cdot \frac{w}{\gamma r_o^3}$	$D_H = \frac{0.288}{\sqrt{B_H}}$
Rocking	$B_R = \frac{3(1-\mu)}{8} \cdot \frac{I_R g}{\gamma r_o^5}$	$D_R = \frac{0.15}{(1+B_R) \sqrt{B_R}}$
Torsion	$B_T = \frac{I_T g}{\gamma r_o^5}$	$D_T = \frac{0.50}{1+2 B_T}$

Table 2. Embedment Damping Factors

Motion	Coefficient S (embedment)
Vertical	$S_V = [1 + 1.9(1-\mu) (H/r_o)] / \sqrt{N_V}$
Horizontal	$S_H = [1 + 1.9(2-\mu) (H/r_o)] / \sqrt{N_H}$
Rocking	$S_R = [1 + 1.7(1-\mu) (H/r_o)] + 0.6(2-\mu) (H/r_o)^3 / \sqrt{N_R}$
Torsion	$S_T = 1.0$



VIBRATING MACHINERY FOUNDATIONS ON SOIL (USING FREQUENCY INDEPENDENT CRITERIA)
**Vertical Damping Coefficients For A Rigid Circular Foundation
Resting On An Elastic Layer Of Thickness T Over A Rigid Base**

T/r_o	∞	4	3	2	1
D_w/D_v	1.00	0.31	0.16	0.09	0.044

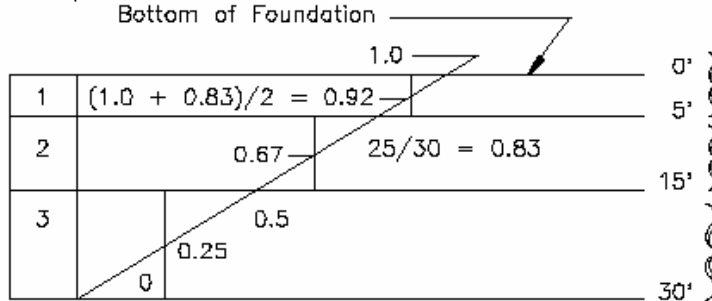
**Horizontal And Rocking Damping Coefficients For A Rigid Circular Foundation
Resting On An Elastic Layer Of Thickness T Over A Rigid Base**

T/r_o	∞	4	3	2	1
D_{Hr}/D_H	1.00	0.66	0.58	0.55	0.52
D_{Rl}/D_R	1.00	0.89	0.85	0.85	0.84



CALCULATION OF EFFECTIVE G VALUE

Example:



Depth = r_0 = 30'

Average values for layer

$\mu = 0.4$ G = 5,000 psi
 $\mu = 0.35$ G = 10,000 psi
 $\mu = 0.3$ G = 15,000 psi

Layer	Wtg Factor	μ	G	(WF x μ)	(WF x G)
1	5 x 0.92 = 4.60	0.4	5,000	1.840	23,000
2	10 x 0.67 = 6.70	0.35	10,000	2.345	67,000
3	15 x 0.25 = 3.75	0.3	15,000	1.125	56,250
Sum	15.05			5.310	146,250

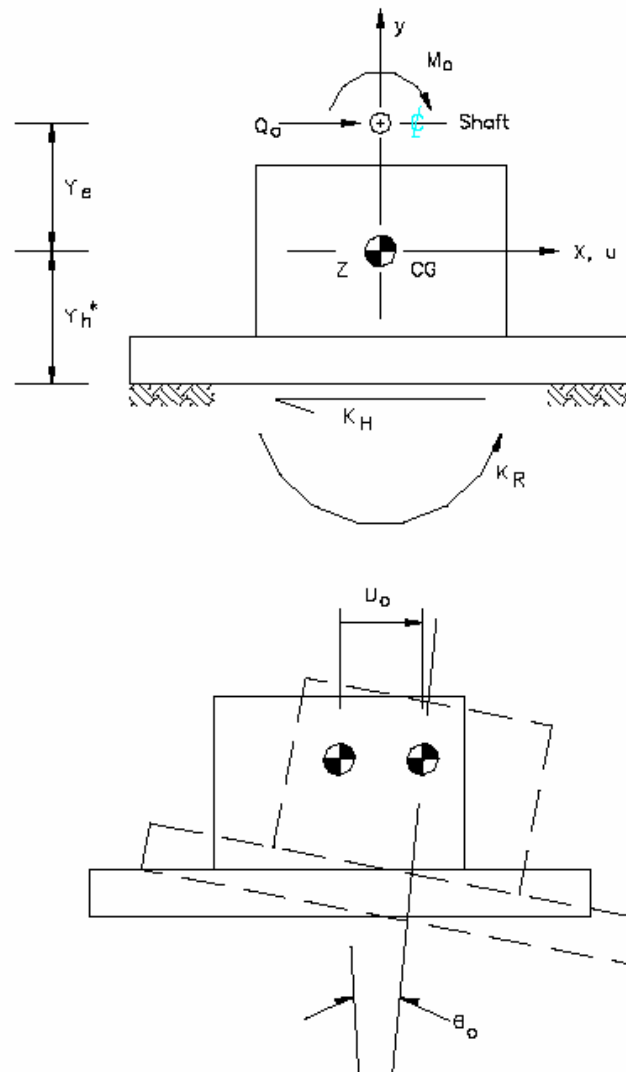
$$\text{Average } \mu = \frac{5.310}{15.05} = 0.35$$

$$\text{Average G} = \frac{146,250}{15.05} = 9,720 \text{ psi}$$



TYPICAL VALUES FOR SOIL PROPERTIES

Description	Allow Bearing (lb / ft ²)	Allow Bearing (kN / m ²)	Soil Weight (lb / ft ³)	Soil Weight (kN / m ³)	Poisson's Ratio (μ)	Shear Modulus G (y)	Shear Modulus G (kN / m ²)
Granite	>10,000	>500	150 – 160	23.6 – 25.1	0.15 – 0.20	(4 - 6) x 10 ⁶	(28 - 41) x 10 ⁶
Limestone	>10,000	>500	145 – 155	22.8 – 24.3	0.16 – 0.22	(2 - 5) x 10 ⁶	(14 - 34) x 10 ⁶
Sandstone	>10,000	>500	145 – 155	22.8 – 24.3	0.17 – 0.24	(1 - 4) x 10 ⁶	(7 - 28) x 10 ⁶
Dense Sand	7,000 - 10,000	350 – 500	115 – 140	18.1 – 22.0	0.28 – 0.34	(10 – 19) x 10 ³	(69 - 131) x 10 ³
Medium Sand	5,000 - 7,000	250 – 350	110 – 130	17.3 – 20.4	0.30 – 0.36	(8 – 15) x 10 ³	(55 - 103) x 10 ³
Loose Sand	3,000 - 5,000	250 – 250	95 – 125	14.5 – 19.6	0.32 – 0.38	(5 – 11) x 10 ³	(34 - 76) x 10 ³
Hard Clay	4,000 - 6,000	200 – 300	125 – 145	19.6 – 22.8	0.38 – 0.41	(11 – 15) x 10 ³	(76 - 103) x 10 ³
Medium Clay	2,000 - 4,000	100 – 200	115 – 135	18.1 – 21.2	0.41 – 0.44	(7 – 11) x 10 ³	(48 - 76) x 10 ³
Soft Clay	1,000 - 2,000	50 – 100	100 – 125	15.7 – 19.6	0.44 – 0.47	(3– 7 x 10 ³	(21 - 48) x 10 ³

COUPLED TRANSLATION -- ROCKING MOTION


CG = Center of Gravity of Combined Machine Foundation System

Q_O, M_O = Unbalanced Force and Moment, Respectively, at Machine Shaft cl.

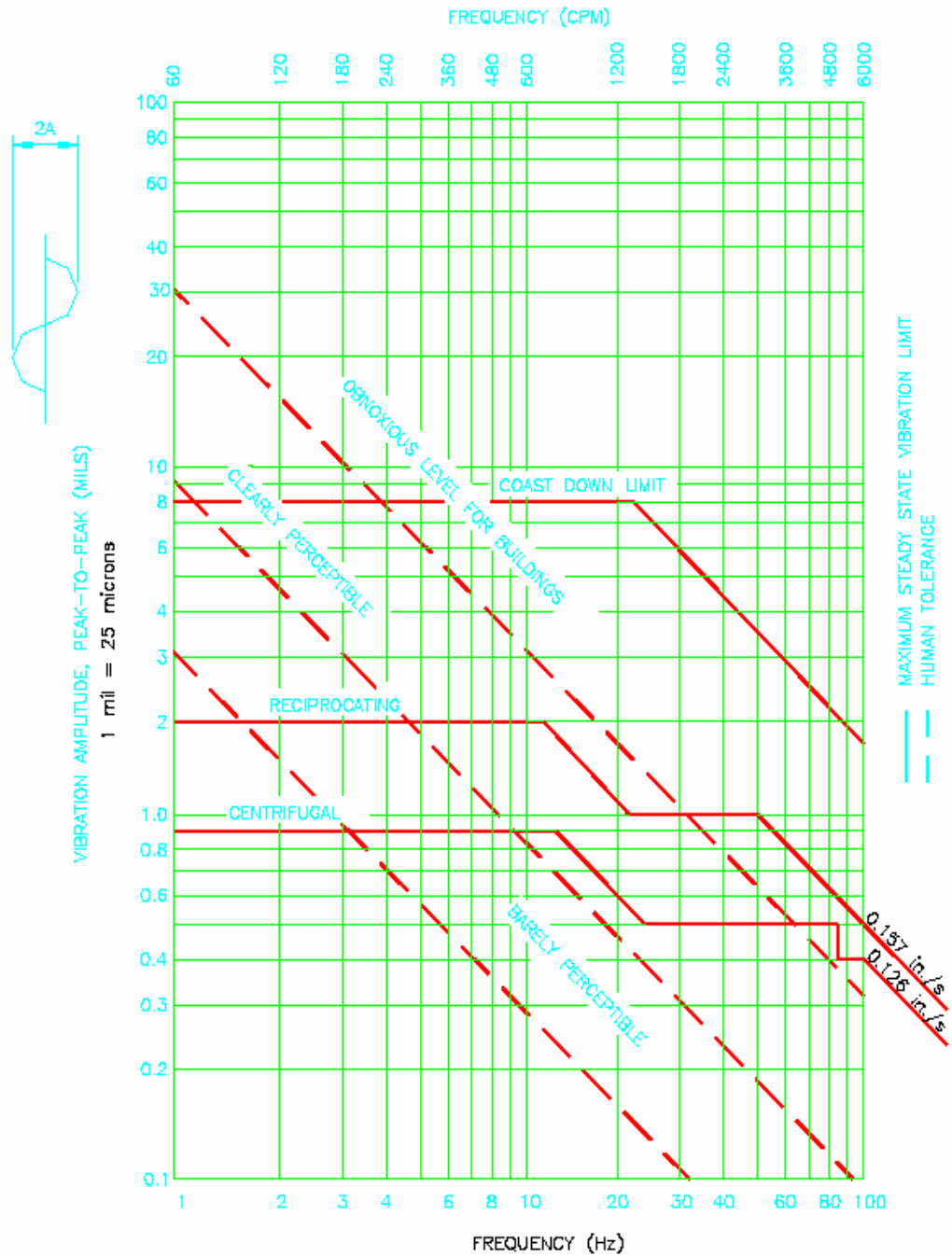
u_O, θ_O = Translation and Rotation Displacements, Respectively, at CG

Note!!! * For Embedded Foundation, y_h Becomes $(y_h + (y_h \frac{H}{3}) (N_H - 1))/N_H$



VIBRATION LIMITS

Vibration Limits (21,38)



**SAMPLE DESIGN 1 :****CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN****DESIGN DATA**

Equipment Data

Compressor

Total Weight	= 18.312 kips
Mass Moment of Inertia axes)	= 5.00 kips-ft-sec ² (same about all 3
Rotor Weight	= 2.317 kips

Motor

Total Weight	= 10.253 kips
Mass Moment of Inertia axes)	= 0.62 kips-ft-sec ² (same about all 3
Rotor Weight	= 2.443 kips

Machine Speed = 1500 rpm

Unbalanced Loads

$$Q_0 = \frac{\text{Rotor Speed}}{6000} \times \text{Rotor Wt.}$$

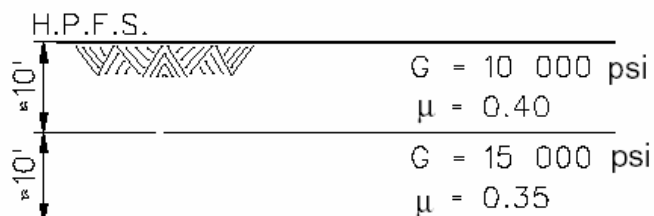
$$\text{Compressor } Q_0 = \frac{1500}{6000} \times 2.317 = 0.58 \text{ kips}$$

$$\text{Motor } Q_0 = \frac{1500}{6000} \times 2.443 = 0.61 \text{ kips}$$

Soil Data

Allowable Soil Bearing = 3.56 ksf
Unit Weight = 0.120 ksf

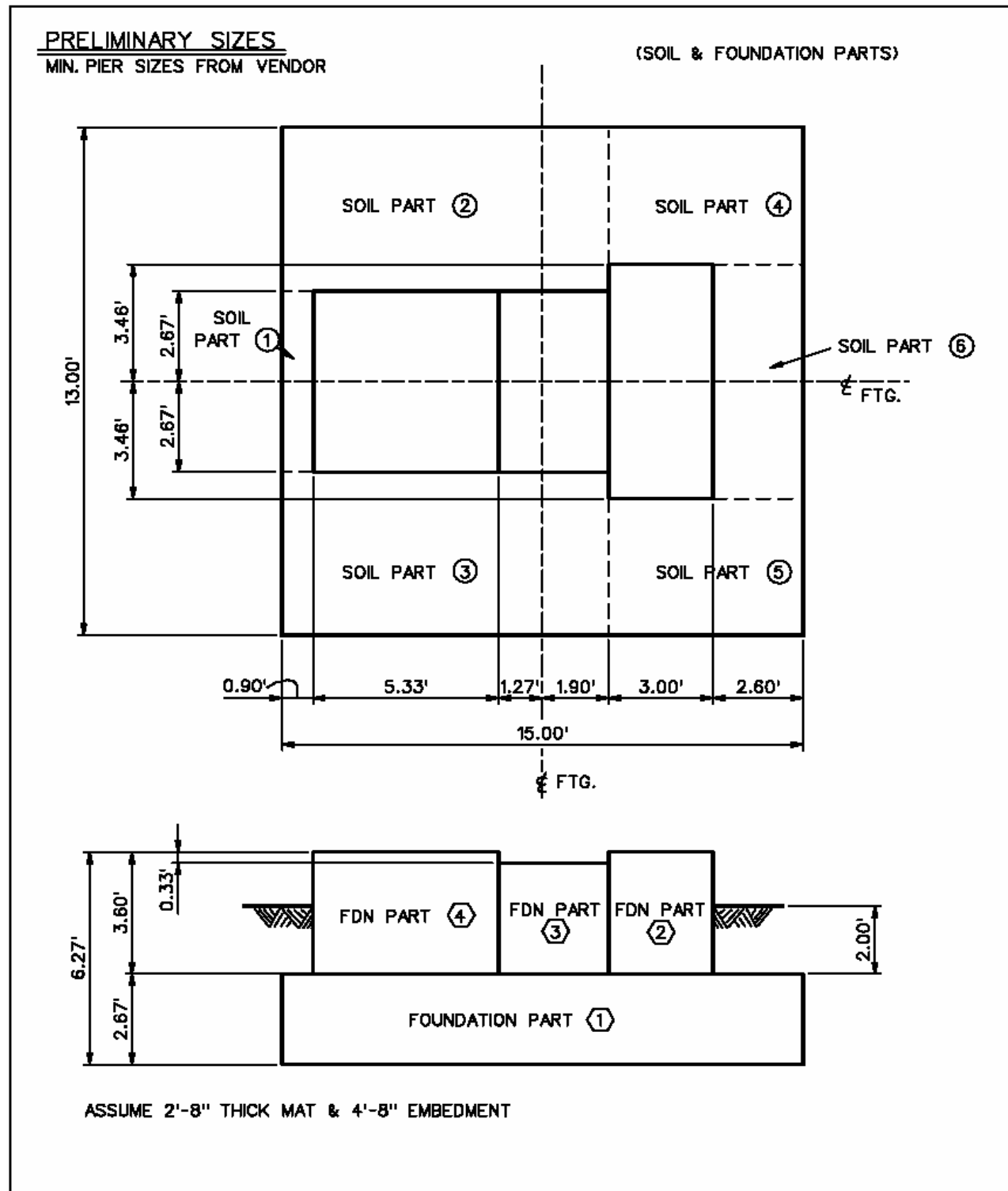
Shear Modulus and Poisson's Ratio vary with Depth





SAMPLE DESIGN 1 :

CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN



**SAMPLE DESIGN 1 :****CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN**PRELIMINARY SIZES (cont'd)

Soil and Foundation Weights

Part No.	Length	Width	Height	Weight
SOIL ①	5.33	0.90	2.00	1.15
SOIL ②	3.83	9.40	2.00	8.64
SOIL ③	3.83	9.40	2.00	8.64
SOIL ④	3.04	5.60	2.00	4.09
SOIL ⑤	3.04	5.60	2.00	4.09
SOIL ⑥	6.92	2.60	2.00	4.32
FDN ①	13.00	15.00	2.67	78.10
FDN ②	6.92	3.00	3.60	11.21
FDN ③	5.33	5.33	3.60	15.34

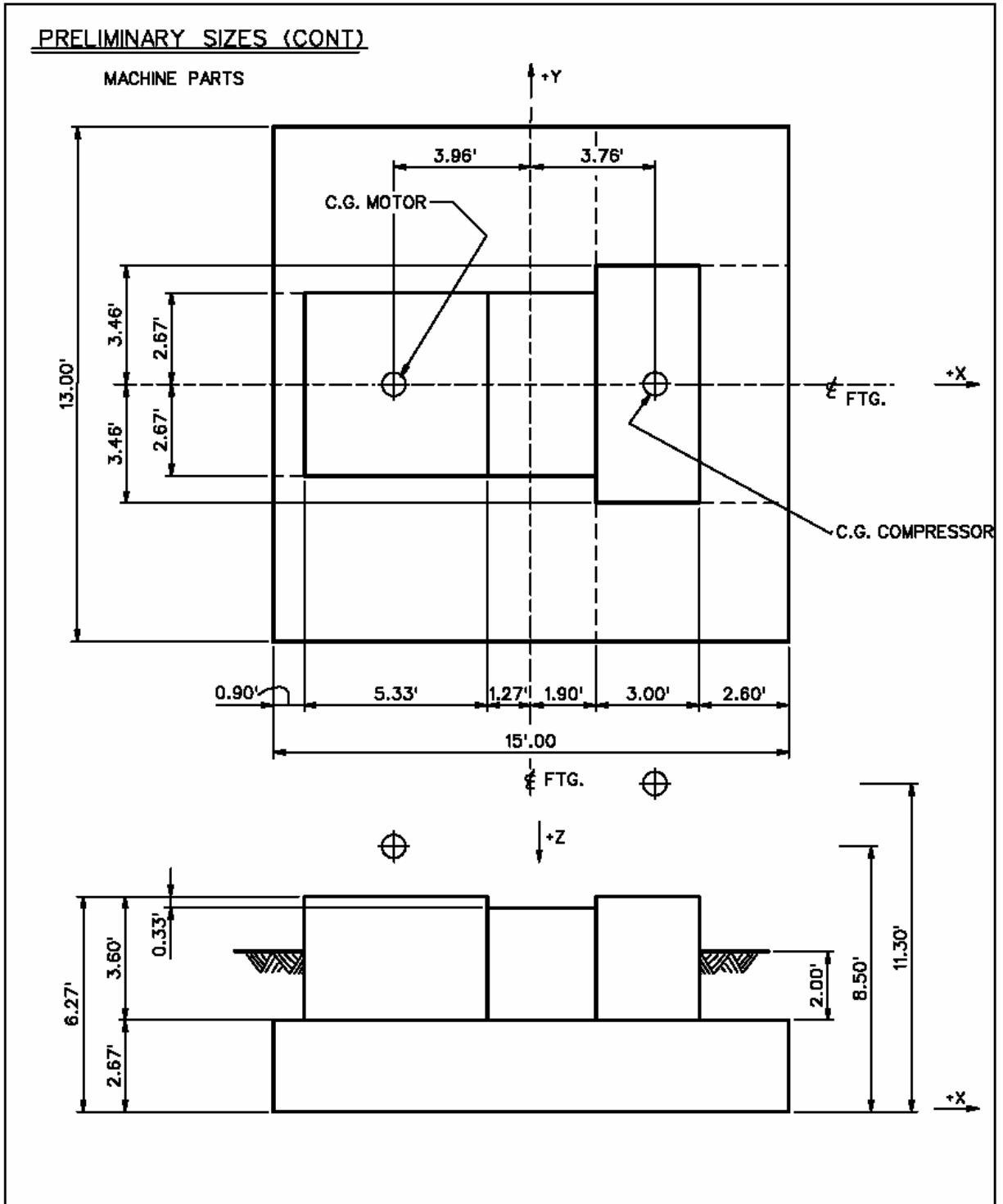
Total Weight = 143.82 + 10.253 + 18.312 = 172.385 kips

$$S.B = \frac{172.385}{15 \times 13} = 0.884 \text{ ksf} < \frac{3.56}{2} \quad \text{O.K.}$$



SAMPLE DESIGN 1 :

CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN





SAMPLE DESIGN 1 :

CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN

PRELIMINARY SIZES (cont'd)

Combined Center of Gravity – (from footing centerline)

Component.	Weight	x	y	z	Wx	Wy	Wz
Compressor	18.312	3.76	0.00	11.30	68.85	0.00	206.90
Motor	10.253	-3.96	0.00	8.50	-40.59	0.00	87.13
FDN ①	78.10	0.00	0.00	1.33	0.00	0.00	103.87
FDN ②	11.21	3.40	0.00	4.47	38.11	0.00	50.11
FDN ③	15.34	-3.94	0.00	4.47	-60.44	0.00	68.57
FDN ④	8.24	0.31	0.00	4.30	2.55	0.00	34.43
SOIL ①	1.15	-7.07	0.00	3.67	-8.13	0.00	4.22
SOIL ②	8.64	-2.80	4.59	3.67	-24.19	39.66	31.71
SOIL ③	8.64	-2.80	-4.59	3.67	-24.19	-39.66	31.71
SOIL ④	4.09	4.70	4.98	3.67	19.22	20.37	15.01
SOIL ⑤	4.09	4.70	-4.98	3.67	19.22	-20.37	15.01
FDN ⑥	4.32	6.20	0.00	3.67	26.78	0.00	15.85

$$\bar{X} = \frac{17.19}{172.385} = 0.0997 \text{ ft eccentricity} = \frac{0.0997}{15.0} = 0.665\%$$

$$\bar{y} = 0 \text{ ft eccentricity} = 0\%$$

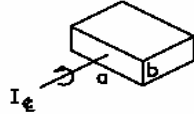
$$\bar{z} = \frac{665.52}{172.385} = 3.861 \text{ ft}$$

Horizontal Eccentricity < 5% in both directions

O.K.

**SAMPLE DESIGN 1 :****CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN****MASS MOMENT OF INERTIA FOR ROCKING ABOUT Y-AXIS**

Axis of rocking is through the combined c.g.



$$I_{CL} = \frac{W}{g} \left(\frac{a^2 + b^2}{12} \right) I_x = I_{CL} + \frac{W}{g} r^2$$

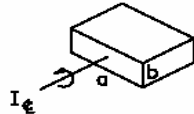
where r = distance from component c.g. to combined c.g.

Component.	Weight	$\frac{1}{12}(a^2 + b^2)$	$I_{CL}g$	r^2	Wr^2
Compressor	18.312	-	161.00	7.44^2	1,013.64
Motor	10.253	-	19.96	-4.46^2	220.74
FDN ①	78.10	$\frac{1}{12}(13^2 + 2.67^2)$	1,146.31	-2.53^2	499.91
FDN ②	11.21	$\frac{1}{12}(6.92^2 + 3.6^2)$	56.84	0.61^2	4.71
FDN ③	15.34	$\frac{1}{12}(5.33^2 + 3.6^2)$	52.88	0.61^2	5.71
FDN ④	8.24	$\frac{1}{12}(5.33^2 + 3.25^2)$	26.75	0.44^2	1.60
SOIL ①	1.15	$\frac{1}{12}(5.33^2 + 2.0^2)$	3.11	-0.19^2	0.04
SOIL ②	8.64	$\frac{1}{12}(3.83^2 + 2.0^2)$	13.44	$-0.14^2 + 4.59^2$	182.34
SOIL ③	8.64	$\frac{1}{12}(3.83^2 + 2.0^2)$	13.44	$-0.19^2 + 4.59^2$	182.34
SOIL ④	4.09	$\frac{1}{12}(3.04^2 + 2.0^2)$	4.51	$-0.19^2 + 4.98^2$	101.58
SOIL ⑤	4.09	$\frac{1}{12}(3.04^2 + 2.0^2)$	4.51	$-0.19^2 + 4.98^2$	101.58
FDN ⑥	4.32	$\frac{1}{12}(6.92^2 + 2.0^2)$	18.68	6.10^2	0.16
Σ	172.385		1,521.43		2,314.35

$$I_{Rx} = \frac{1521.43 + 2314.35}{32.2} = 119.12 \text{ kips} - \text{ft} - \text{sec}^2$$

**SAMPLE DESIGN 1 :****CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN****MASS MOMENT OF INERTIA FOR ROCKING ABOUT Y-AXIS**

Axis of rocking is through the combined c.g.



$$I_{CL} = \frac{W}{g} \left(\frac{a^2 + b^2}{12} \right) I_x = I_{CL} + \frac{W}{g} r^2$$

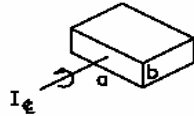
where r = distance from component c.g. to combined c.g.

Component.	Weight	$\frac{1}{12}(a^2 + b^2)$	$I_{CL}g$	r^2	Wr^2
Compressor	18.312	-	161.00	$2.44^2 + 3.66^2$	1,258.94
Motor	10.253	-	19.96	$-4.64^2 + 4.06^2$	389.75
FDN ①	78.10	$\frac{1}{12}(15.0^2 + 2.67^2)$	1,510.77	$-2.53^2 + 0.10^2$	500.69
FDN ②	11.21	$\frac{1}{12}(3.0^2 + 3.6^2)$	20.51	$0.21^2 + 0.10^2$	126.25
FDN 3 ③	15.34	$\frac{1}{12}(5.33^2 + 3.6^2)$	52.88	$0.61^2 + (-4.04)^2$	256.08
FDN ④	8.24	$\frac{1}{12}(3.17^2 + 3.25^2)$	14.15	$0.44^2 + 0.22^2$	1.99
SOIL ①	1.15	$\frac{1}{12}(0.9^2 + 2.0^2)$	0.46	$-0.19^2 + (-7.15)^2$	58.83
SOIL ②	8.64	$\frac{1}{12}(9.4^2 + 2.0^2)$	66.50	$-0.19^2 + (-2.90)^2$	72.97
SOIL ③	8.64	$\frac{1}{12}(9.4^2 + 2.0^2)$	66.50	$-0.19^2 + (-2.90)^2$	72.97
SOIL ④	4.09	$\frac{1}{12}(5.6^2 + 2.0^2)$	12.05	$-0.19^2 + 4.60^2$	86.69
SOIL ⑤	4.09	$\frac{1}{12}(5.6^2 + 2.0^2)$	12.05	$-0.19^2 + 4.60^2$	86.69
FDN ⑥	4.32	$\frac{1}{12}(2.6^2 + 2.0^2)$	3.87	$-0.19 + 6.10^2$	160.90
Σ	172.385		1,940.70		3,072.75

$$IR_z = \frac{1940.70 + 3072.75}{32.2} = 155.70 \text{ kips} - \text{ft} - \text{sec}^2$$

**SAMPLE DESIGN 1 :****CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN****MASS MOMENT OF INERTIA FOR ROCKING ABOUT Y-AXIS**

Axis of rocking is through the combined c.g.



$$I_{CL} = \frac{W}{g} \left(\frac{a^2 + b^2}{12} \right) I_x = I_{CL} + \frac{W}{g} r^2$$

where r = distance from component c.g. to combined c.g.

Component.	Weight	$\frac{1}{12}(a^2 + b^2)$	$I_{CL}g$	r^2	Wr^2
Compressor	18.312	-	161.00	3.66^2	245.30
Motor	10.253	-	19.96	-4.06^2	169.01
FDN ①	78.10	$\frac{1}{12}(13.0^2 + 15.0^2)$	2,564.28	0.10^2	0.78
FDN ②	11.21	$\frac{1}{12}(6.92^2 + 3.0^2)$	53.14	3.30^2	122.08
FDN 3 ③	15.34	$\frac{1}{12}(5.33^2 + 5.33^2)$	72.63	4.04^2	250.37
FDN ④	8.24	$\frac{1}{12}(5.33^2 + 3.17^2)$	26.41	0.21^2	0.68
SOIL ①	1.15	$\frac{1}{12}(3.83^2 + 0.9^2)$	2.80	-7.17^2	59.12
SOIL ②	8.64	$\frac{1}{12}(3.83^2 + 9.4^2)$	74.18	$-2.9^2 + 4.59^2$	254.69
SOIL ③	8.64	$\frac{1}{12}(3.83^2 + 9.4^2)$	74.18	$-2.9^2 + (-4.59)^2$	254.69
SOIL ④	4.09	$\frac{1}{12}(3.04^2 + 5.6^2)$	13.84	$-4.6^2 + 4.98^2$	187.98
SOIL ⑤	4.09	$\frac{1}{12}(3.04^2 + 5.6^2)$	13.84	$-4.6^2 + (-4.98)^2$	187.98
FDN ⑥	4.32	$\frac{1}{12}(6.92^2 + 2.6^2)$	19.67	6.10^2	160.75
Σ	172.385		3,095.93		1,893.43

$$IR_z = \frac{3095.93 + 1893.43}{32.2} = 154.95 \text{ kips} - \text{ft} - \text{sec}^2$$

**SAMPLE DESIGN 1 :****CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN****EQUIVALENT RADII**

Aspect Ratio of Footing = $13/15 = 0.87 > 1/3 < 3$

Use equivalent circular footing

Translation (Vertical and Horizontal)

$$r_o = \sqrt{\frac{bl}{\pi}} = \sqrt{\frac{15(13)}{\pi}} = 7.88 \text{ ft}$$

Rocking about x-axis

$$r_o = \left(\frac{bl^3}{3\pi} \right)^{\frac{1}{4}} = \left(\frac{15(13)^3}{3\pi} \right)^{\frac{1}{4}} = 7.69 \text{ ft}$$

Rocking about y-axis

$$r_o = \left(\frac{bl^3}{3\pi} \right)^{\frac{1}{4}} = \left(\frac{15(13)^3}{3\pi} \right)^{\frac{1}{4}} = 8.26 \text{ ft}$$

Rotating about z-axis

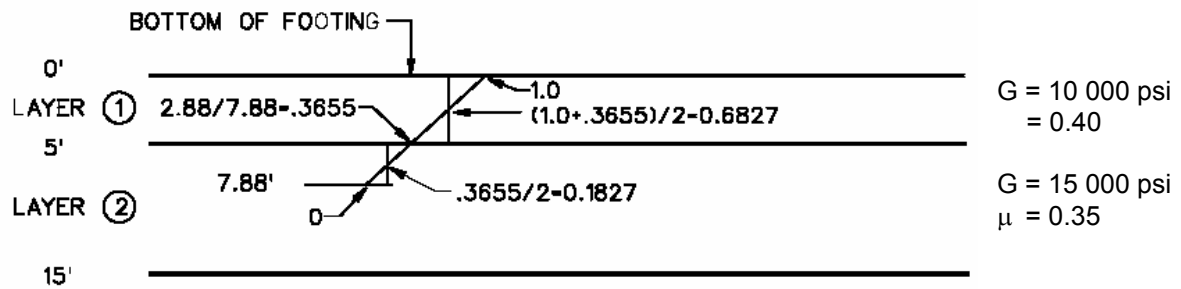
$$r_o = \left(\frac{bl(b^2 + l^2)}{6\pi} \right)^{\frac{1}{4}} = \left(\frac{15(13)(13^2 + 15^2)}{6\pi} \right)^{\frac{1}{4}} = 7.94 \text{ ft}$$

For a rigorous analysis, the shear modulus and Poisson's ratio of the soil must be calculated for each mode of vibration.



**SAMPLE DESIGN 1 :
 CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN**

SHEAR MODULUS AND POISSON'S RATIO FOR TRANSLATION ($r_0 = 7.88$ ft)



Layer	Wtg. Factor	μ	G	WF x μ	WF x G
1	$5 \times 0.6827 = 3.41$	0.40	10,000.00	1.365	34100
2	$2.88 \times 0.183 = 0.53$	0.35	15,000.00	0.184	7890
Σ	3.94			1.534	41990

$$\mu_T = \frac{1.534}{3.940} = 0.3894 \quad G_T = \frac{41990}{3.94} = 10657 \text{ psi} = 1534 \text{ ksf}$$

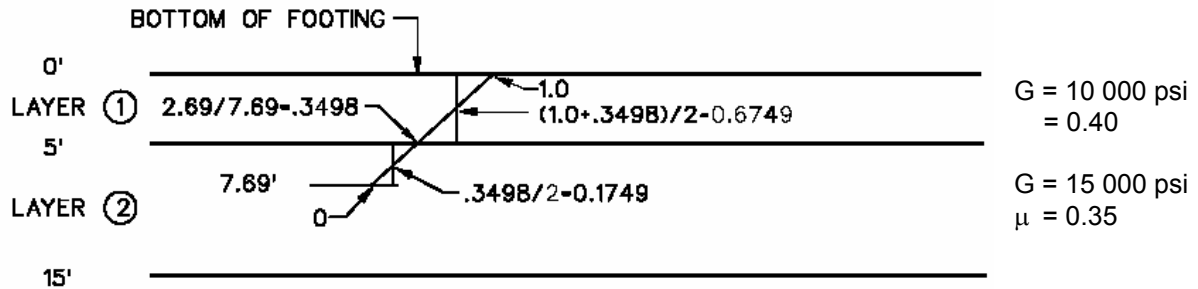
$$\mu_V = \mu_H = 0.3894$$

$$G_V = G_H = 10657 \text{ psi} = 1534 \text{ ksf}$$



SAMPLE DESIGN 1 :
CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN

SHEAR MODULUS AND POISSON'S RATIO FOR ROCKING ABOUT X-AXIS ($r_x = 7.69$ ft)



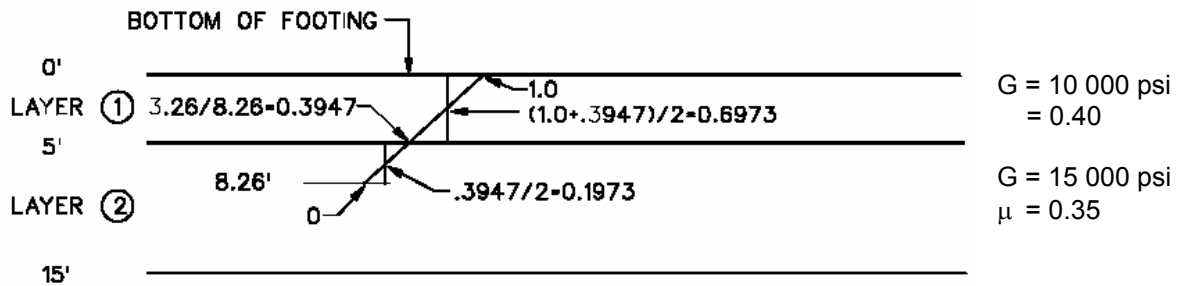
Layer	Wtg. Factor	μ	G	WF x μ	WF x G
1	$5 \times 0.675 = 3.375$	0.40	10,000.00	1.35	33750
2	$2.69 \times 0.175 = 0.47$	0.35	15,000.00	0.165	7050
Σ	3.845			1.515	40800

$$\mu_{Rx} = \frac{1.515}{3.845} = 0.394 \quad G_{Rx} = \frac{40800}{3.845} = 10611.2 \text{ psi} = 1528 \text{ ksf}$$



SAMPLE DESIGN 1 :
CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN

SHEAR MODULUS AND POISSON'S RATIO FOR ROCKING ABOUT Y-AXIS ($r_0 = 8.26$ ft)



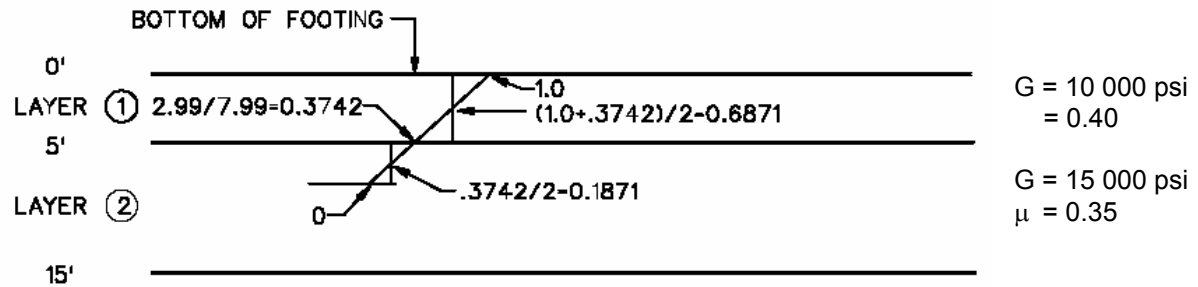
Layer	Wtg. Factor	μ	G	WF x μ	WF x G
1	$5 \times 0.6973 = 3.487$	0.40	10,000.00	1.395	34870
2	$3.26 \times 0.1973 = 0.640$	0.35	15,000.00	0.225	9648
Σ	4.130			1.620	44518

$$\mu_{Ry} = \frac{1.620}{4.130} = 0.392 \quad G_{Ry} = \frac{44518}{4.130} = 10779.2 \text{ psi} = 1552 \text{ ksf}$$



SAMPLE DESIGN 1 :
CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN

SHEAR MODULUS AND POISSON'S RATIO FOR ROCKING ABOUT Z-AXIS ($r_0 = 7.99$ ft)



Layer	Wtg. Factor	μ	G	WF x μ	WF x G
1	$5 \times 0.6871 = 3.44$	0.40	10,000.00	1.370	34400
2	$2.99 \times 0.1871 = 0.56$	0.35	15,000.00	0.196	8392
Σ	0.8742			1.566	42792

$$\mu_{Rz} = \frac{1.566}{4.0} = 0.39 \quad G_{Rz} = \frac{42792}{4.0} = 10698 \text{ psi} = 1541 \text{ ksf}$$

Minimum Footing Thickness

$$t = 0.02 \left(\frac{G_{\max} L^4}{b} \right)^{\frac{1}{3}} = 0.02 \left(\frac{1540.5 \left(\frac{1000}{144} \right) 3.83^4}{13} \right)^{\frac{1}{3}} = 1.12 \text{ ft} < 2.67 \text{ ft} \quad \text{O.K.}$$

**SAMPLE DESIGN 1 :****CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN****EMBEDMENT STIFFNESS FACTORS (N)**

Embedment Depth = 4.67 ft Use 2/3 of actual depth for analysis -- $H = 4.67 \left(\frac{2}{3} \right) = 3.11$ ft

$$N_V = 1 + 0.6(1 - \mu_V) \frac{H}{r_0} = 1 + 0.6(1 - 0.3894) \frac{3.11}{7.88} = 1.145$$

$$N_H = 1 + 0.55(2 - \mu_H) \frac{H}{r_0} = 1 + 0.55(2 - 0.3894) \frac{3.11}{7.88} = 1.350$$

$$N_{R_x} = 1 + 1.2(1 - \mu_{R_x}) \frac{H}{r_0} + 0.2(2 - \mu_{R_x}) \left(\frac{H}{r_0} \right)^3 = 1 + 1.2(1 - 0.3940) \frac{3.11}{7.69} + 0.2(2 - 0.3940) \left(\frac{3.11}{7.69} \right)^3 = 1.315$$

$$N_{R_y} = 1 + 1.2(1 - \mu_{R_y}) \frac{H}{r_0} + 0.2(2 - \mu_{R_y}) \left(\frac{H}{r_0} \right)^3 = 1 + 1.2(1 - 0.3920) \frac{3.11}{8.26} + 0.2(2 - 0.3920) \left(\frac{3.11}{8.26} \right)^3 = 1.292$$

$$N_{R_z} = 1.0$$

Spring Constants

Footings aspect ratio is within the range of 1/3 to 3, therefore use equivalent radii equations

$$K_V = \frac{4G_V r_0}{1 + \mu_V} \times N_V = \frac{4(1534)(7.88)}{1 - 0.3894} \times 1.145 = 9.067 \times 10^4 \frac{\text{kips}}{\text{ft}}$$

$$K_H = \frac{32(1 - \mu_H)G_H r_0}{7 + 8\mu_H} = \frac{32(1 - 0.3894)(1534)(7.88)}{7 - 8(0.3894)} \times 1.350 = 8.207 \times 10^4 \frac{\text{kips}}{\text{ft}}$$

$$K_{R_x} = \frac{8G_{R_x} r_0^3}{3(1 + \mu_{R_x})} \times N_{R_x} = \frac{8(1528)(7.69)^3}{3(1 - 0.3940)} \times 1.315 = 4.020 \times 10^6 \frac{\text{kips} \cdot \text{ft}}{\text{rad}}$$

$$K_{R_y} = \frac{8G_{R_y} r_0^3}{3(1 + \mu_{R_y})} \times N_{R_y} = \frac{8(1552)(8.26)^3}{3(1 - 0.392)} \times 1.292 = 4.960 \times 10^6 \frac{\text{kips} \cdot \text{ft}}{\text{rad}}$$

$$K_{R_z} = \frac{16G_{R_z} r_0^3}{3} \times N_{R_z} = \frac{16(1541)(7.99)^3}{3} \times 1.0 = 4.192 \times 10^6 \frac{\text{kips} \cdot \text{ft}}{\text{rad}}$$

**SAMPLE DESIGN 1 :****CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN****EMBEDMENT DAMPING FACTORS (S)**

$$S_V = \frac{1 + 1.9(1 - \mu_V) \left(\frac{H}{r_0}\right)}{\sqrt{N_V}} = \frac{1 + 1.9(1 - 0.3894) \left(\frac{3.11}{7.88}\right)}{\sqrt{1.145}} = 1.362$$

$$S_H = \frac{1 + 1.9(2 - \mu_H) \left(\frac{H}{r_0}\right)}{\sqrt{N_H}} = \frac{1 + 1.9(2 - 0.3894) \left(\frac{3.11}{7.88}\right)}{\sqrt{1.350}} = 1.901$$

$$S_{R_x} = \frac{1 + 0.7(1 - \mu_{R_x}) \left(\frac{H}{r_0}\right) + 0.6(2 - \mu_{R_x}) \left(\frac{H}{r_0}\right)^3}{\sqrt{N_{R_x}}} = \frac{1 + 0.7(1 - 0.394) \left(\frac{3.11}{7.69}\right) + 0.6(2 - 0.394) \left(\frac{3.11}{7.69}\right)^3}{\sqrt{1.315}} = 1.079$$

$$S_{R_y} = \frac{1 + 0.7(1 - \mu_{R_y}) \left(\frac{H}{r_0}\right) + 0.6(2 - \mu_{R_y}) \left(\frac{H}{r_0}\right)^3}{\sqrt{N_{R_y}}} = \frac{1 + 0.7(1 - 0.392) \left(\frac{3.11}{8.26}\right) + 0.6(2 - 0.392) \left(\frac{3.11}{8.26}\right)^3}{\sqrt{1.292}} = 1.067$$

$$S_{R_z} = 1.0$$

MASS RATIOS

Vertical - -

$$B_V = \frac{1 - \mu_V}{4} \left(\frac{W}{\gamma r_0^3} \right) = \frac{1 - 0.3894}{4} \left[\frac{172.385}{0.12(7.88)^3} \right] = 0.4482$$

Horizontal - -

$$B_H = \frac{7 - 8\mu_H}{32(1 - \mu_H)} \left(\frac{W}{\gamma r_0^3} \right) = \frac{7 - 8(0.3894)}{32(1 - 0.3894)} \left(\frac{172.385}{0.12(7.88)^3} \right) = 0.5837$$

Rocking about x-axis

$$B_{R_x} = \frac{3(1 - \mu_{R_x})}{8} \left(\frac{I_{R_x} g}{\gamma r_0^3} \right) = \frac{3(1 - 0.394)}{8} \left(\frac{119.12(32.2)}{0.12(7.69)^3} \right) = 0.2699$$

Rocking about y-axis

$$B_{R_y} = \frac{3(1 - \mu_{R_y})}{8} \left(\frac{I_{R_y} g}{\gamma r_0^3} \right) = \frac{3(1 - 0.392)}{8} \left(\frac{155.70(32.2)}{0.12(8.26)^3} \right) = 0.2477$$

Rotation about z-axis

$$B_z = \left(\frac{I_{R_z} g}{\gamma r_0^3} \right) = \left(\frac{154.95(32.2)}{0.12(7.99)^3} \right) = 1.2768$$



**SAMPLE DESIGN 1 :
 CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN**

MASS RATIOS (cont'd)

B_{Rx} & B_{Ry} need to be corrected for small rocking masses

$$B_{Rx} < 1 \rightarrow n_{Rx} = \frac{1.219}{B_{Rx}^{0.169}} = 1.5190$$

$$B_{Ry} < 1 \rightarrow n_{Ry} = \frac{1.219}{B_{Ry}^{0.169}} = 1.3419$$

Corrected Mass Ratios - -

B_V	=	0.4482
B_H	=	0.5837
B_{Rx}	=	$0.2699(1.5190) = 0.4099$
B_{Ry}	=	$0.2477(1.5419) = 0.3819$
B_{Rz}	=	1.2768

Mass moments of inertia also require correction for small rocking masses.

Effective Mass Moments of Inertia –

$$I_{Rx} = I_{Rx} \cdot n_{Rx} = 119.12(1.5190) = 180.94 \text{ kips} \cdot \text{ft} \cdot \text{sec}^2$$

$$I_{Ry} = I_{Ry} \cdot n_{Ry} = 155.70(1.5419) = 240.07 \text{ kips} \cdot \text{ft} \cdot \text{sec}^2$$

$$I_{Rz} = 154.95$$

DAMPING RATIOS

Vertical

$$D_V = \left(\frac{0.425}{\sqrt{B_V}} + 0.05 \right) S_V = \left(\frac{0.425}{\sqrt{0.4482}} + 0.05 \right) .363 = 0.9334 < 0.95 \quad (\text{Max. allow. vert. damping})$$

Horizontal

O.K.

$$D_H = \left(\frac{0.288}{\sqrt{B_H}} + 0.05 \right) S_H = \left(\frac{0.288}{\sqrt{0.5837}} + 0.05 \right) 1.901 = 0.8117 < 0.60 \quad (\text{Max. allow. horiz. damping})$$

USE $D_H = 0.600$

**SAMPLE DESIGN 1 :
 CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN**

DAMPING RATIOS (cont'd)

Rocking about x-axis

$$D_{Rx} = \left[\frac{0.15}{(1+B_{Rx})\sqrt{B_{Rx}}} + 0.05 \right] S_{Rx} = \left[\frac{0.15}{(1+0.4099)\sqrt{0.4099}} + 0.05 \right] 1.078 = 0.233 < 0.40$$

(Max. allow. rocking damping = 0.40)

O.K.

Rocking about y-axis

$$D_{Ry} = \left[\frac{0.15}{(1+B_{Ry})\sqrt{B_{Ry}}} + 0.05 \right] S_{Ry} = \left[\frac{0.15}{(1+0.3819)\sqrt{0.3819}} + 0.05 \right] 1.066 = 0.2406 < 0.40$$

(Max. allow. rocking damping = 0.40)

O.K.

Rocking about z-axis

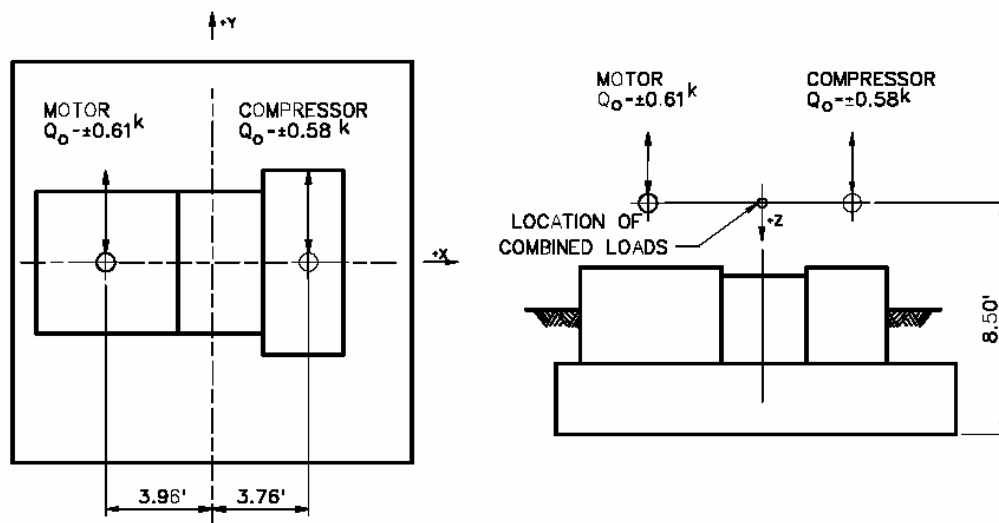
$$D_{Rz} = \frac{0.5}{1+2B_{Rz}} + 0.05 = \frac{0.5}{1+2(1.2768)} + 0.05 = 0.1907 < 0.20$$

(Max. allow. rocking damping = 0.40)

O.K.

LOADING

Load locations shown below:



**SAMPLE DESIGN 1 :****CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN**LOADING (cont'd)

Since the rotors are operating at the same speed, the dynamic loads must be considered in-phase and 180° out-of-phase

Consider only the in-phase condition for this example

Equivalent loads

$$\begin{array}{rclclcl} F_x & = & 0 & M_x & = & 0 \\ F_y & = & 1.19 \text{ kips} & M_y & = & -0.235 \text{ kips-ft} \\ F_z & = & 1.19 \text{ kips} & M_z & = & -0.235 \text{ kips-ft} \end{array}$$

Equivalent loads located 8.50 ft above centroid of footing.

COUPLED ANALYSIS

Translation in x and Rocking about y

Because of embedment, y_h must be modified. (y_h = vertical distance from combined c.g. to B.O.F.)

$$y_h = \frac{Y_h K_H + (y_h - \frac{H}{3}) K_H (N_H - 1)}{K_H N_H} = \frac{3.861(8.207 \times 10^4) + (3.861 - \frac{4.67}{3})(8.207 \times 10^4)(1.35 - 1)}{240.07} = 3.457$$

$$MASS = \frac{W}{q} = \frac{172.385}{32.2} = 5.354$$

Coupled Frequencies

$$\omega_i^4 - B\omega_i^2 - C = 0$$

$$B = \frac{K_H}{M} + \frac{(K_{RY} + K_H Y_H^2)}{I_{RY}} = \frac{8.207 \times 10^4}{5.354} + \frac{(4.96 \times 10^6 + (8.207 \times 10^4)(3.457^2))}{240.07} = 4.007 \times 10^4$$

$$C = \frac{K_H K_{RY}}{M I_{RY}} = \frac{8.207 \times 10^4 (4.96 \times 10^6)}{5.354 (240.07)} = 3.167 \times 10^8$$

$$\omega_i^4 = (4.007 \times 10^4) \omega_i^2 + 3.167 \times 10^8 = 0$$

$$\omega_i^2 = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A} = \frac{4.007 \times 10^4 \pm \sqrt{(4.007 \times 10^4)^2 - 4(3.167 \times 10^8)}}{2}$$

$$\omega_1^2 = 1.083 \times 10^4 + \left(\frac{\text{rad}}{\text{sec}} \right)^2 \quad -\omega_1 = 104.067 \frac{\text{rad}}{\text{sec}} = 993.76 \text{ rpm}$$

$$\omega_2^2 = 2.9238 \times 10^4 + \left(\frac{\text{rad}}{\text{sec}} \right)^2 \quad -\omega_2 = 170.99 \frac{\text{rad}}{\text{sec}} = 1632.8 \text{ rpm}$$

**SAMPLE DESIGN 1 :****CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN**COUPLED ANALYSIS -- TRANS. X AND ROCKING Y (cont'd)

Modal Coordinates

$$U_i = \frac{K_H x Y_h}{K_H - M \omega_i^2}$$

$$U_1 = \frac{8.207 \times 10^4 (3.4574)}{8.207 \times 10^4 - 5.354 (1.083 \times 10^4)} = 11.78$$

$$U_1 = \frac{8.207 \times 10^4 (3.4574)}{8.207 \times 10^4 - 5.354 (2.924 \times 10^4)} = -3.8097$$

Generalized Masses

$$M_i = M U_i^2 + I_{RY}$$

$$M_1 = 5.354 (11.780)^2 + 240.07 = 983.04$$

$$M_2 = 5.354 (-3.8097)^2 + 240.07 = 317.78$$

Generalized Forces

$$F_i = Q_0 U_i + M_0 + Q_0 y_e + Q_0 h'$$

For translation in the x direction and rocking about the y axis

$$Q_0 = 0 \quad Q_0' = 1.19 \text{ kips} \quad M_0 = M_y = M_z = -0.235 \text{ kips} \cdot \text{ft} \quad h' = -0.1 \text{ ft}$$

$$F_1 = -0.354 \text{ kips}$$

$$F_2 = -0.354 \text{ kips}$$

Note: h' = horizontal component of load location minus horizontal eccentricity

Equivalent Modal Damping Ratios

$$D_{ix}^* = \frac{U_i^2 K_H D_H + K_{Ry} D_{Ry}}{U_i^2 K_H + K_{Ry}}$$

$$D_{1x}^* = \frac{(11.78)^2 8.207 \times 10^4 (0.600) + 4.96 \times 10^6 (0.2393)}{(11.78)^2 8.207 \times 10^4 + 4.96 \times 10^6} = 0.491$$

$$D_{2x}^* = \frac{(-3.8097)^2 8.207 \times 10^4 (0.600) + 4.96 \times 10^6 (0.2393)}{(-3.8097)^2 8.207 \times 10^4 + 4.96 \times 10^6} = 0.3087$$

**SAMPLE DESIGN 1 :****CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN**COUPLED ANALYSIS -- TRANS -X AND ROCKING - Y (cont'd)

Steady-State Response Values

$$q_i = \frac{F_i}{M_i} \left[(\omega_i^2 - \Omega^2)^2 + (2D_i^* \omega_i \Omega)^2 \right]^{-\frac{1}{2}}$$

$$\Omega = 1500 \text{ rpm} = \frac{1}{60} \times 2\pi = 157.08 \frac{\text{rad}}{\text{sec}}$$

$$q_1 = \frac{-0.354}{983.04} \left[(1.083 \times 10^4 - 157.08^2)^2 + (2(0.491)104.067(157.08))^2 \right]^{-\frac{1}{2}} = -1.698 \times 10^{-8}$$

$$q_2 = \frac{-0.354}{317.78} \left[(2.9238 \times 10^4 - 157.08^2)^2 + (2(0.3087)170.99(157.08))^2 \right]^{-\frac{1}{2}} = -6.4768 \times 10^{-8}$$

Phase Angles

$$\alpha_i = \arctan \left(\frac{2D_i^* \omega_i \Omega}{\omega_i^2 - \Omega^2} \right)$$

$$\alpha_1 = \arctan \left[\frac{2(0.491)104.067(157.08)}{1.083 \times 10^4 - 157.08^2} \right] = -49.224 \text{ deg}$$

$$\alpha_2 = \arctan \left[\frac{2(0.3087)170.99(157.08)}{2.9238 \times 10^4 - 157.08^2} \right] = 74.61 \text{ deg}$$

Translation at c.g.

$$\begin{aligned} U_{0x} &= \left[(q_1 U_1 \sin \alpha_1 + q_2 U_2 \sin \alpha_2)^2 + (q_1 U_1 \cos \alpha_1 + q_2 U_2 \cos \alpha_2)^2 \right]^{\frac{1}{2}} \\ &= \left[(-1.698 \times 10^{-8} (11.780) \sin(-49.224) + (-6.4768 \times 10^{-8}) (-3.8097) \sin(74.61))^2 \right. \\ &\quad \left. + (-1.698 \times 10^{-8} (11.78) \cos(-49.224) + (-6.4768 \times 10^{-8}) (-3.8097) \cos(74.61))^2 \right]^{\frac{1}{2}} \\ &= 3.9477 \times 10^{-7} \text{ ft} \Rightarrow 0.0048 \text{ mils} \end{aligned}$$

Rotation at c.g.

$$\begin{aligned} \theta_0 &= \left[(q_1 \sin \alpha_1 + q_2 \sin \alpha_2)^2 + (q_1 \cos \alpha_1 + q_2 \cos \alpha_2)^2 \right]^{\frac{1}{2}} \\ \theta_{0x} &= \left[(-1.698 \times 10^{-8} \sin(-49.224) + (-6.4768 \times 10^{-8}) (-3.8097) \sin(74.61))^2 \right. \\ &\quad \left. + (-1.698 \times 10^{-8} \cos(-49.224) + (-6.4768 \times 10^{-8}) \cos(74.61))^2 \right]^{\frac{1}{2}} \\ &= 2.5234 \times 10^{-7} \text{ rad} = 1.445 \times 10^{-5} \text{ deg} \end{aligned}$$

**SAMPLE DESIGN 1 :****CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN**COUPLED ANALYSISTranslation in y and Rocking about x

Coupled Frequencies

$$\omega_i^4 - B\omega_i^2 - C = 0$$

$$B = \frac{K_H}{M} + \frac{(K_{Rx} + K_H Y_H^2)}{I_{Rx}} = \frac{8.207 \times 10^4}{5.354} + \frac{[4.02 \times 10^6 + (8.207 \times 10^4)(3.4574^2)]}{180.94} = 4.296 \times 10^4$$

$$C = \frac{K_H K_{Rx}}{M I_{Rx}} = \frac{8.207 \times 10^4 (4.02 \times 10^6)}{5.354 (180.94)} = 3.4056 \times 10^8$$

$$\omega_i^4 - (4.296 \times 10^4) \omega_i^2 + 3.4056 \times 10^8 = 0$$

$$\omega_i^2 = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A} = \frac{4.296 \times 10^4 \pm \sqrt{(4.296 \times 10^4)^2 - 4(3.4056 \times 10^8)}}{2}$$

$$\omega_1^2 = 1.0487 \times 10^4 \left(\frac{\text{rad}}{\text{sec}} \right)^2 \quad -\omega_1 = 102.41 \frac{\text{rad}}{\text{sec}} = 977.94 \text{ rpm}$$

$$\omega_2^2 = 3.2472 \times 10^4 \left(\frac{\text{rad}}{\text{sec}} \right)^2 \quad -\omega_2 = 180.2 \frac{\text{rad}}{\text{sec}} = 1720.8 \text{ rpm}$$

Modal Coordinates

$$U_i = \frac{K_H K_h}{K_H - M \omega_i^2}$$

$$U_1 = \frac{8.207 \times 10^4 (3.4574)}{8.207 \times 10^4 - 5.354 (1.0487 \times 10^4)} = 10.946$$

$$U_2 = \frac{8.207 \times 10^4 (3.4574)}{8.207 \times 10^4 - 5.354 (3.2472 \times 10^4)} = -3.0914$$

Generalized Masses

$$M_i = M U_i^2 + I_{Rx}$$

$$M_1 = 5.354 (10.946)^2 + 180.94 = 822.429$$

$$M_2 = 5.354 (-3.0914)^2 + 180.94 = 232.107$$

**SAMPLE DESIGN 1 :****CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN****COUPLED ANALYSIS TRANS -Y AND ROCKING X (cont'd)****Generalized Forces**

$$F_i = Q_0 U_i + M_0 + Q_0 y_e + Q_0 \dot{h}$$

For translation in y direction and rocking about the x axis

$$Q_0 = F_y = 1.19 \text{ kips} \quad M_0 = M_x = 0 \quad \dot{h} = 0$$

$$Q_0 = F_z = 1.19 \text{ kips}$$

$$y_e = 8.5 - 3.861 = 4.639 \text{ ft} \quad (8.5 = \text{vert. coord. of load, } 3.861 = \text{vert. coord. of c.g.})$$

$$F_1 = 1.19(10.946) + 1.19(4.639) = 18.546 \text{ kips}$$

$$F_2 = 1.19(-3.0914) + 1.19(4.639) = 1.8416 \text{ kips}$$

Equivalent Modal Damping Ratios

$$D_{iy}^* = \frac{U_i^2 K_H D_H + K_{Ry} D_{Ry}}{U_i^2 K_H + K_{Ry}}$$

$$D_{1y}^* = \frac{(10.946)^2 8.207 \times 10^{-4} (0.600) + 4.96 \times 10^{-6} (0.2406)}{(10.9061)^2 8.207 \times 10^{-4} + 4.960 \times 10^{-6}} = 0.47$$

$$D_{2y}^* = \frac{(-3.0914)^2 8.207 \times 10^{-4} (0.600) + 4.960 \times 10^{-6} (0.2406)}{(-3.0914)^2 8.207 \times 10^{-4} + 4.960 \times 10^{-6}} = 0.2896$$

Steady State Response Values

$$q_i = \frac{F_i}{M_i} \left[(\omega_i^2 - \Omega^2)^2 + (2D_i^* \omega_i \Omega)^2 \right]^{-\frac{1}{2}}$$

$$q_1 = \frac{18.546}{822.429} \left[(1.0487 \times 10^4 - 157.08^2)^2 + (2(0.47)102.41(157.08))^2 \right]^{-\frac{1}{2}} = 1.0875 \times 10^{-6}$$

$$q_2 = \frac{1.8416}{232.107} \left[(3.2472 \times 10^4 - 157.08^2)^2 + (2(0.2896)180.2(157.08))^2 \right]^{-\frac{1}{2}} = 4.3704 \times 10^{-7}$$

Phase Angles

$$\alpha_i = \arctan \left(\frac{2D_i^* \omega_i \Omega}{\omega_i^2 - \Omega^2} \right)$$

$$\alpha_1 = \arctan \left[\frac{2(0.47)102.41(157.08)}{1.0487 \times 10^4 - 157.08^2} \right] = -46.825 \text{ deg}$$

$$\alpha_2 = \arctan \left[\frac{2(0.2896)180.2(157.08)}{3.2472 \times 10^4 - 157.08^2} \right] = 64.6 \text{ deg}$$



SAMPLE DESIGN 1 :

CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN

COUPLED ANALYSIS -- TRANS - Y AND ROCKING X (cont'd)

Translation at c.g.

$$\begin{aligned}
 U_{oy} &= \left[(q_1 U_1 \sin \alpha_1 + q_2 U_2 \sin \alpha_2)^2 + (q_1 U_1 \cos \alpha_1 + q_2 U_2 \cos \alpha_2)^2 \right]^{\frac{1}{2}} \\
 &= \left[(1.0875 \times 10^{-6} (10.946) \sin(-46.825) + (4.3704 \times 10^{-7}) (-3.0914) \sin(64.6))^2 \right. \\
 &\quad \left. + (1.0875 \times 10^{-6} (10.946) \cos(-46.825) + (4.3704 \times 10^{-7}) (-3.0914) \cos(64.6))^2 \right]^{\frac{1}{2}} \\
 &= 1.2461 \times 10^{-5} \text{ ft } 0.1495 \text{ mils}
 \end{aligned}$$

Rotation at c.g.

$$\begin{aligned}
 \theta_{oy} &= \left[(q_1 \sin \alpha_1 + q_2 \sin \alpha_2)^2 + (q_1 \cos \alpha_1 + q_2 \cos \alpha_2)^2 \right]^{\frac{1}{2}} \\
 &= \left[(1.0875 \times 10^{-6} \sin(-46.825) + (4.3704 \times 10^{-7}) \sin(64.6))^2 \right. \\
 &\quad \left. + (1.0875 \times 10^{-6} \cos(-46.825) + (4.3704 \times 10^{-7}) \cos(64.6))^2 \right]^{\frac{1}{2}} \\
 &= 1.013 \times 10^{-7} \text{ rad} = 5.8041 \times 10^{-5} \text{ deg}
 \end{aligned}$$

UNCOUPLED ANALYSIS

Undamped Natural Frequencies

Vertical

$$f_f = \frac{1}{2\pi} \sqrt{\frac{K_v}{M}} = \frac{1}{2\pi} \sqrt{\frac{9.067 \times 10^4}{5.354}} = 20.71 \text{ Hz} = 1242.7 \text{ rpm}$$

Rotation about z

$$f_f = \frac{1}{2\pi} \sqrt{\frac{K_{Rz}}{I_{Rz}}} = \frac{1}{2\pi} \sqrt{\frac{4.192 \times 10^6}{154.95}} = 26.178 \text{ Hz} = 1570.67 \text{ rpm}$$

Dynamic Magnification Factors

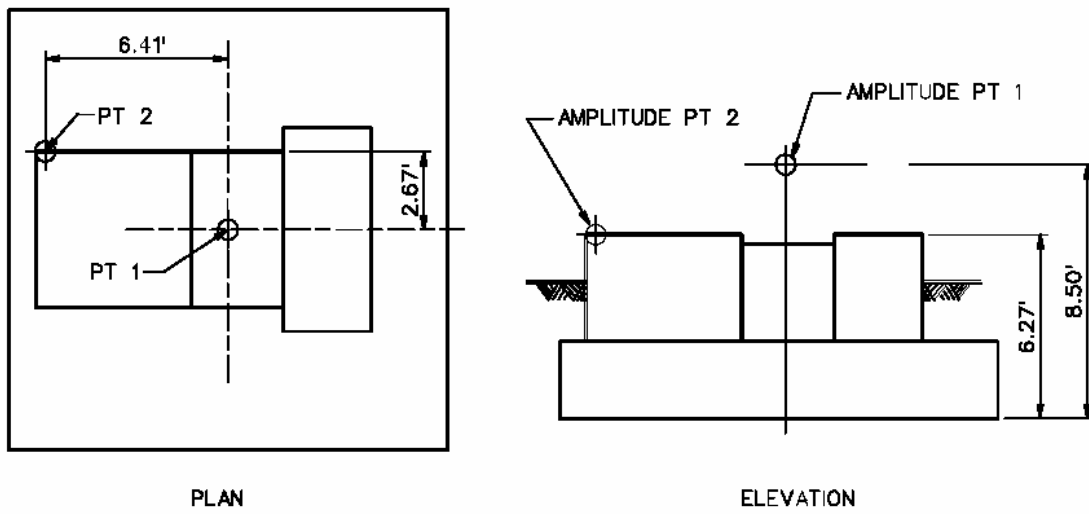
$$M = \frac{1}{\sqrt{\left(1 - \left(\frac{f}{f_f}\right)^2\right)^2 + \left(2D \frac{f}{f_f}\right)^2}}$$

$$M_y = \frac{1}{\sqrt{\left(1 - \left(\frac{1500}{1242.7}\right)^2\right)^2 + \left(2(0.9334)\left(\frac{1500}{1242.7}\right)\right)^2}} = 0.4349$$

$$M_{Rz} = \frac{1}{\sqrt{\left(1 - \left(\frac{1500}{1570.67}\right)^2\right)^2 + \left(2(0.1907)\left(\frac{1500}{1570.67}\right)\right)^2}} = 2.6687$$

**SAMPLE DESIGN 1 :****CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN**AMPLITUDE POINTS

CHECK AMPLITUDES AT POINTS SHOWN BELOW

AMPLITUDES OF VIBRATION

Pt. 1 (0, 0, 8.5)

z-Contribution from Vertical Vibration

$$S = \frac{Q_0}{K_V} = \frac{1.190}{9.067 \times 10^4} = 1.3125 \times 10^{-5}$$

$$A = |M_V S| = 0.4349(1.3125 \times 10^{-5}) = 5.7081 \times 10^{-6} \text{ ft}$$

$$\text{Peak - Peak amplitude} = 2(12)(1000)5.7081 \times 10^{-6} = 0.137 \text{ mils}$$

x-Contribution from Rotation about z-axis

$$\phi = \frac{Q_0 h + T_0}{K_{RZ}} = \frac{-0.24}{4.192 \times 10^6} = -5.725 \times 10^{-8}$$

$$A = |M_T \phi h'| \text{ but } h' = 0 \text{ therefore, peak - peak amplitude} = 0.000 \text{ mils}$$



SAMPLE DESIGN 1 :

CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN

AMPLITUDES OF VIBRATION (cont'd)

Pt. 1 (0, 0, 8.5)

y-Contribution from Rotation about z-axis

$$\phi = \frac{1.19(-0.0997) - 0.24}{4.192 \times 10^{-6}} = -8.5554 \times 10^{-8}$$

$$A = 2.6687(-8.5554 \times 10^{-8})(0 - 0.0997) = 2.2763 \times 10^{-8} \text{ ft}$$

$$\text{peak - peak amplitude} = 2A = 0.00054 \text{ mils}$$

x-Contribution from Translation in x and Rocking about y-axis

$$A = |U_{0x} + \theta_{0x} d'| = 3.9477 \times 10^{-7} + (2.5234 \times 10^{-7})(8.5 - 3.861) = 15.402 \times 10^{-7} \text{ ft}$$

$$\text{peak - peak amplitude} = 2A = 0.0369 \text{ mils}$$

z-Contribution from Translation in x and Rocking about y-axis

$$A = |\theta_{0x} d'| = 3.7010 \times 10^{-8} \times (0 - 0.0997) = 3.6899 \times 10^{-9} \text{ ft}$$

$$\text{peak - peak amplitude} = 2A = 0.0001 \text{ mils}$$

y-Contribution from Translation in y and Rocking about x-axis

$$A = |U_{0y} + \theta_{0y} d'| = 1.2461 \times 10^{-5} + (10.013 \times 10^{-7})(8.5 - 3.861) = 1.7106 \times 10^{-5} \text{ ft}$$

$$\text{peak - peak amplitude} = 2A = 0.4105 \text{ mils}$$

z-Contribution from Translation in y & Rocking About x-Axis

$$A = |\theta_{0y} x d'| \text{ but } d' = 0 \text{ therefore peak - peak amplitude} = 0$$

Pt. 2 (-6.41, 2.67, 6.27)

z-Contribution from Vertical Vibration

$$\text{peak-peak amplitude} = 0.137 \text{ mils (ref. pt. 1)}$$

x-Contribution from Rotation about the z-axis

$$A = |M_T \phi h'| = 2.6687(-8.5554 \times 10^{-8})(2.67 - 0) = 6.096 \times 10^{-7} \text{ ft}$$

$$\text{peak - peak amplitude} = 2A = 0.0146 \text{ mils}$$



**SAMPLE DESIGN 1 :
CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN**

AMPLITUDES OF VIBRATIONS (cont'd)

Pt. 2 (-6.41, 2.67, 6.27)

y-Contribution from Rotation about z-axis

$$A = |M_T \phi h'| = 2.6687(-8.5554 \times 10^{-8})(-6.41 - 0.0997) = 1.4863 \times 10^{-6} \text{ ft}$$

peak - peak amplitude = $2A = 0.03567$ mils

x-Contribution from Translation in x and Rocking about y-axis

$$A = |U_{0x} + \theta_{0x} d'| = 3.9477 \times 10^{-7} + (2.5234 \times 10^{-7})(6.27 - 3.861) = 10.0266 \times 10^{-7} \text{ ft}$$

peak - peak amplitude = $2A = 0.0241$ mils

z-Contribution from Translation in x and Rocking about y-axis

$$A = |\theta_{0x} d'| = 3.7010 \times 10^{-8} x(-6.41 - 0.0997) = 2.4092 \times 10^{-7} \text{ ft}$$

peak - peak amplitude = $2A = 0.0058$ mils

y-Contribution from Translation in y and Rocking about x-axis

$$A = |U_{0y} + \theta_{0y} d'| = 1.2491 \times 10^{-5} + (10.013 \times 10^{-7})(6.27 - 3.861) = 1.4903 \times 10^{-5} \text{ ft}$$

peak - peak amplitude = $2A = 0.3576$ mils

z-Contribution from Translation in y and Rocking about x-axis

$$A = |\theta_{0y} d'| = 10.013 \times 10^{-7} x(2.67 - 0) = 2.674 \times 10^{-6} \text{ ft}$$

peak - peak amplitude = $2A = 0.0642$ mils



**SAMPLE DESIGN 1 :
CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN**

AMPLITUDE SUMMARY

Amplitude Pt. 1

Mode of Vibration	Peak to Peak Amplitudes (Mils)		
	x	y	z
Vertical	--	--	0.1370
Rotation about z –axis	0.0000	0.0005	--
Trans. -x and Rocking	0.0369	--	0.0001
-y			
Trans. -y and Rocking	--	0.4105	0.0000
-x			
Total	0.0369	0.41104	0.1371

Amplitude Pt. 2

Mode of Vibration	Peak to Peak Amplitudes (Mils)		
	x	y	z
Vertical	--	--	0.1370
Rotation about z –axis	0.0146	0.03567	--
Trans. -x and Rocking	0.0241	--	0.0058
-y			



**SAMPLE DESIGN 1 :
 CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN**

COAST DOWN ANALYSIS

The largest contributing mode of vibration is horizontal translation in the y-direction and rocking about the x-axis

The frequencies of the coupled mode are

$$\omega_1 = 102.41 \frac{\text{rad}}{\text{sec}} = 997.94 \text{ rpm}$$

$$\omega_1 = 180.2 \frac{\text{rad}}{\text{sec}} = 1720.8 \text{ rpm}$$

ω_2 will not be critical since it is greater than the frequency of excitation ($\Omega = 1500 \text{ rpm}$)

Machine Resonant Speed

$$f^* = \frac{F_f}{\sqrt{1 - 2D^2}} = \frac{997.94}{\sqrt{1 - 2(0.47)^2}} = 1335.75 \text{ rpm} = 139.9 \frac{\text{rad}}{\text{sec}}$$

Coast Down Unbalanced Forces

$$Q_0^* = \left(\frac{f^*}{f} \right)^2 Q_0 = \left(\frac{1335.75}{1500} \right)^2 Q_0 = 0.7929 Q_0$$

$$\begin{array}{llll} F_{x_s}^* & = & 0 & M_{x_s}^* & = & 0 \\ F_{y_s}^* & = & 1.0194 \text{ kips} & M_{y_s}^* & = & -0.2011 \text{ kips} - \text{ft} \\ F_{z_s}^* & = & 1.0194 \text{ kips} & M_{z_s}^* & = & -0.2011 \text{ kips} - \text{ft} \end{array}$$

Coupled Analysis -- Trans. - x and Rocking - y

Coupled frequencies, modal coordinates, generalized masses and equivalent modal damping do not change.



**SAMPLE DESIGN 1 :
 CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN**

COAST DOWN ANALYSIS (cont'd)

Coupled Analysis -- Trans. -x and Rocking -y

Generalized Forces $F_i = Q_0 U_i + M_0 + Q_0 y_e + Q_0 h'$

$$F_1 = M_y = -0.3031$$

$$F_2 = M_y = -0.3031$$

Steady State Response ($\Omega^* = 145.38 \text{ rad/sec}$)

$$q_1 = \frac{-0.3031}{983.04} \left[(1.083 \times 10^4 - 145.38^2)^2 + (2(0.491)(104.067)(145.38))^2 \right]^{-\frac{1}{2}} = -1.7052 \times 10^{-8}$$

$$q_2 = \frac{-0.3031}{317.78} \left[(2.9238 \times 10^4 - 145.38^2)^2 + (2(0.3087)(170.99)(145.38))^2 \right]^{-\frac{1}{2}} = -5.496 \times 10^{-8}$$

Phase Angles

$$\alpha_1 = \arctan \left[\frac{2(0.491)(104.067)(145.38)}{1.083 \times 10^4 - 145.38^2} \right] = -55.253 \text{ deg}$$

$$\alpha_2 = \arctan \left[\frac{2(0.3087)(170.99)(145.38)}{2.9238 - 145.38^2} \right] = 62.17 \text{ deg}$$

Translation at c.g.

$$\begin{aligned} U_{ox} &= \left[(-1.7052 \times 10^{-8} (11.78) \sin(-55.253) + (-5.496 \times 10^{-8})(-3.8097) \sin(62.17))^2 \right. \\ &\quad \left. + (-1.7052 \times 10^{-8} (11.78) \cos(-55.253) + (-5.496 \times 10^{-8})(-3.8097) \cos(62.17))^2 \right]^{\frac{1}{2}} \\ &= 3.5413 \times 10^{-7} \text{ ft} \Rightarrow 0.00425 \text{ mils} \end{aligned}$$

Rotation at c.g.

$$\begin{aligned} \theta_{ox} &= \left[(-1.7052 \times 10^{-8} \sin(-55.253) + (-5.496 \times 10^{-8}) \sin(62.17))^2 \right. \\ &\quad \left. + (-1.7052 \times 10^{-8} \cos(-55.253) + (-5.496 \times 10^{-8}) \cos(62.17))^2 \right]^{\frac{1}{2}} \\ &= 4.9478 \times 10^{-8} \text{ rad} \Rightarrow 2.8349 \times 10^{-6} \text{ degrees} \end{aligned}$$

**SAMPLE DESIGN 1 :****CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN**COAST DOWN ANALYSIS (cont'd)

Coupled Analysis -- Trans. -y and Rocking -x

Generalized Forces

$$F_1 = 1.0194(10.946) + 1.0194(4.639) = 15.887$$

$$F_2 = 1.0194(-3.0914) + 1.0194(4.639) = 1.5776$$

Steady State Response ($\Omega^* = 145.38$ rad/sec)

$$q_1 = \frac{15.887}{822.429} \left[(1.0487 \times 10^4 - 145.38^2)^2 + (2(0.47)(102.41)(145.38))^2 \right]^{-\frac{1}{2}}$$

$$= 1.0985 \times 10^{-6}$$

$$q_2 = \frac{1.5776}{232.107} \left[(3.2472 \times 10^4 - 145.38^2)^2 + (2(0.2896)(180.2)(145.38))^2 \right]^{-\frac{1}{2}}$$

$$= 3.5885 \times 10^{-7}$$

Phase Angles

$$\alpha_1 = \arctan \left[\frac{2(0.47)(102.41)(145.38)}{1.0487 \times 10^4 - 145.38^2} \right] = -52.734 \text{ deg}$$

$$\alpha_2 = \arctan \left[\frac{2(0.2896)(180.2)(145.38)}{3.2472 \times 10^4 - 145.38^2} \right] = 53.235 \text{ deg}$$

Translation at c.g.

$$U_{y_0} = \left[(1.0985 \times 10^{-6})(10.946) \sin(-52.734) + (3.5885 \times 10^{-7})(-3.0914) \sin(53.235) \right]^2$$

$$+ (1.0985 \times 10^{-6})(10.946) \cos(-52.734) + (3.5885 \times 10^{-7})(-3.0914) \cos(53.235) \right]^2$$

$$= 1.0915 \times 10^{-5} \text{ ft} \Rightarrow 0.1309 \text{ mils}$$

$$\theta_{y_0} = \left[(1.0985 \times 10^{-6} \sin(-52.734) + (3.5885 \times 10^{-7}) \sin(53.235) \right]^2$$

$$+ (1.0985 \times 10^{-6} \cos(-52.734) + (3.5885 \times 10^{-7}) \cos(53.235) \right]^2$$

$$= 1.0576 \times 10^{-6} \text{ rad} \Rightarrow 6.06 \times 10^{-5} \text{ degrees}$$



**SAMPLE DESIGN 1 :
 CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN**

COAST DOWN ANALYSIS (cont'd)

Uncoupled Analysis

Undamped Natural Frequencies as Before

Magnification Factors

$$M_y = \frac{1}{\sqrt{\left(1 - \left(\frac{1335.75}{1242.7}\right)^2\right)^2 + \left(2(0.9334)\left(\frac{1335.75}{1242.7}\right)\right)^2}} = 0.4968$$

$$M_{RZ} = \frac{1}{\sqrt{\left(1 - \left(\frac{1335.75}{1570.67}\right)^2\right)^2 + \left(2(0.1907)\left(\frac{1335.75}{1570.67}\right)\right)^2}} = 2.3453$$

Amplitudes of Vibration

Pt. 1 (0, 0, 8.5)

z-Contribution from Vertical Vibration

$$S = \frac{1.019}{9.067 \times 10^4} = 1.1238 \times 10^{-5}$$

$$A = 0.4968(1.1238 \times 10^{-5}) = 5.5830 \times 10^{-6} \text{ ft}$$

$$\text{Peak - Peak Amplitude} = 2A = 0.1339 \text{ mils}$$

x-Contribution from Rotation about z-axis = 0.0000 mils

y-Contribution from Rotation about z-axis

$$\phi = \frac{1.019(-0.0997) - 0.206}{4.192 \times 10^6} = -7.337 \times 10^{-8}$$

$$A = 2.3453(-7.337 \times 10^{-8})(0 - 0.0997) = 1.7157 \times 10^{-8} \text{ ft}$$

$$\text{Peak - Peak Amplitude} = 2A = 0.0004 \text{ mils}$$



**SAMPLE DESIGN 1 :
CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN**

COAST DOWN ANALYSIS (cont'd)

Amplitudes of Vibration (Pt. 1)

x-Contribution from Translation in x and Rocking about y-axis

$$A = \left| 3.5413 \times 10^{-7} + 4.9478 \times 10^{-8} (8.5 - 3.861) \right| = 5.8366 \times 10^{-7} \text{ ft}$$

$$\text{Peak - Peak Amplitude} = 2A = 0.0140 \text{ mils}$$

z-Contribution from Translation in x and Rocking about y-axis

$$A = \left| 3.1965 \times 10^{-8} \times (0 - 0.0997) \right| = 3.1869 \times 10^{-9}$$

$$\text{Peak - Peak Amplitude} = 2A = 0.0001 \text{ mils}$$

y-Contribution from Translation in y and Rocking about x-axis

$$A = \left| 1.0915 \times 10^{-5} + 1.0576 \times 10^{-6} (8.5 - 3.861) \right| = 1.5821 \times 10^{-5} \text{ ft}$$

$$\text{Peak - Peak Amplitude} = 2A = 0.3797 \text{ mils}$$

z-Contribution from Translation in y and Rocking about x-axis

$$\text{Peak - Peak Amplitude} = 0.000 \text{ because } d = 0$$

Pt. 2 (-6.41, 2.67, 6.27)

z-Contribution from Vertical Vibration = 0.1339 mils

x-Contribution from Rotation about z-axis

$$A = \left| 2.3453(-7.337 \times 10^{-8})(2.67) \right| = 4.5944 \times 10^{-7}$$

$$\text{Peak - Peak Amplitude} = 2A = 0.0110 \text{ mils}$$



**SAMPLE DESIGN 1 :
CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN**

COAST DOWN ANALYSIS (cont'd)

Amplitudes of Vibration (Pt. 2)

y-Contribution from Rotation about z-axis

$$A = \left| 2.3453(-7.337 \times 10^{-8})(-6.41 - 0.0997) \right| = 1.1202 \times 10^{-6}$$

Peak - Peak Amplitude = $2A = 0.0269$ mils

x-Contribution from Translation in x and Rocking about y-axis

$$A = \left| 3.5413 \times 10^{-7} + 4.9478 \times 10^{-8}(6.27 - 3.861) \right| = 4.7332 \times 10^{-7}$$

Peak - Peak Amplitude = $2A = 0.0113$ mils

z-Contribution from Translation in x and Rocking about y-axis

$$A = \left| (3.1965 \times 10^{-8})(-6.41 - 0.0997) \right| = 2.0808 \times 10^{-7}$$

Peak - Peak Amplitude = $2A = 0.0050$ mils

y-Contribution from Translation in y and Rocking about x-axis

$$A = \left| 1.0915 \times 10^{-5} + 1.0576 \times 10^{-6}(6.27 - 3.861) \right| = 1.3463 \times 10^{-5}$$

Peak - Peak Amplitude = $2A = 0.3231$ mils

z-Contribution from Translation in y and Rocking about x-axis

$$A = \left| (1.0576 \times 10^{-6})(2.67 - 0) \right| = 2.824 \times 10^{-6}$$

Peak - Peak Amplitude = $2A = 0.0677$ mils



**SAMPLE DESIGN 1 :
CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN**

COAST DOWN ANALYSIS (cont'd)

Amplitude Summary

Amplitude Pt. 1 (0, 0, 8.5)

Mode of Vibration	Peak to Peak Amplitudes (Mils)		
	X	y	z
Vertical	--	--	0.1339
Rotation about z -axis	0.0000	0.0004	--
Trans. -x and Rocking -y	0.0140	--	0.0001

Amplitude Pt. 2 (-6.41, 2.67, 6.27)

Mode of Vibration	Peak to Peak Amplitudes (Mils)		
	X	y	z
Vertical	--	--	0.1339
Rotation about z -axis	0.0110	0.0269	--
Trans. -x and Rocking -y	0.0113	--	0.0050

Note: For an actual design, these calculations need to be repeated for the unbalanced forces 180° out-of-phase.



**SAMPLE DESIGN 1 :
CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN**

COMPUTER OUTPUT. SVAP

FLUOR DANIEL

SVAP (For Review Only, 9/16/94)

OPCON 000.215.1233

Date: 10/19/94
Contract: OPCON
By:
Page: 1

Machine Forces:

Machine Data-	Location of Forces-
Centrifugal machine.	N/S coordinate = N 0.000 ft
Speed = 1500 rpm	E/W coordinate = E 0.000 ft
	Elevation = 108.500 ft
Direction-	Primary-
N/S force	1.19 K
E/W force	0.00 K
Vertical force	1.19 K
Moment about N/S axis	0.24 K-ft
Moment about E/W axis	0.00 K-ft
Torsional moment	0.24 K-ft

Machine Parts:

Part name:	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)	Weight: (K)	Mass moments: (K-sec2-ft)			
Compressor	N 0.000	E 3.760	111.300	18.310	5	5	5	
Motor	N 0.000	W 3.960	108.500	10.250	1	1	1	

Amplitude Point Locations:

Name:	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)
Point 1	N 0.000	E 0.000	108.500
Point 2	N 2.670	W 6.410	106.270



**SAMPLE DESIGN 1 :
CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN**

COMPUTER OUTPUT. SVAP. (cont'd)

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Foundation Data:

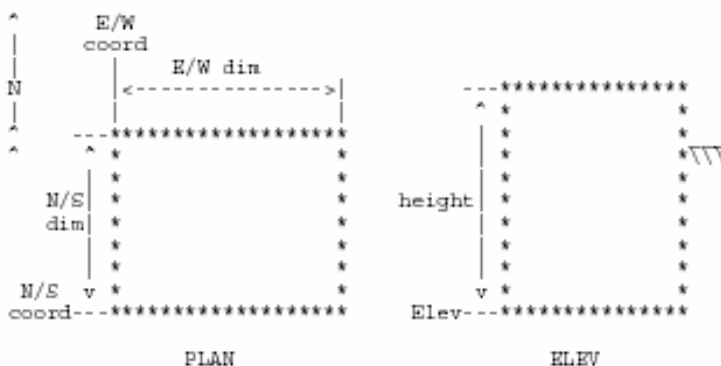
- * Foundation geometry will be checked by the analysis.
- * Group stiffness and damping will be computed from soil layers.
- * The mat will be located as specified.
- * Embedment effects will be computed.

Grade elevation	=	104.67 ft
Effective embedment height	=	4.67 ft
Backfill density	=	120.00 lb/ft3
Poisson's ratio	=	0.40
Stiffness variation	=	1.00
Damping variation	=	1.00
Internal damping	=	5.00 %

Soil Layer Data

Top elev: (ft)	Thickness: (ft)	Density: (lb/ft3)	Shear modulus: (ksf)	Poisson's ratio:
104.670	10.000	120.000	1440.000	0.400
94.670	10.000	120.000	2160.000	0.350

Typical Pier & Mat dimensioning:





**SAMPLE DESIGN 1 :
CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN**

COMPUTER OUTPUT. SVAP. (cont'd)

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Mat Data:

Name:	N/S dim: (ft)	E/W dim: (ft)	Height: (ft)	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)
Part 1	13.000	15.000	2.670	S 6.500	W 7.500	100.000

Pier Data:

Name:	N/S dim: (ft)	E/W dim: (ft)	Height: (ft)	N/S coord: (ft)	E/W coord: (ft)	Elev: (ft)
Part 2	6.920	3.000	3.600	S 3.460	E 1.900	102.670
Part 3	5.330	3.170	3.270	S 2.670	W 1.270	102.670
Part 4	5.330	5.330	3.600	S 2.670	W 6.600	102.700

Results of mass computation:

Center of gravity:		Weight & Mass:	
N/S coordinate =	S 0.000 ft	Total weight =	172.54 K
E/W coordinate =	E 0.098 ft	Machine weight =	28.56 K
elevation =	103.865 ft	Mass =	5.36 K-sec2/ft
		Mass ratio =	5.04 to 1
Footing data:		Mass moments @ CG:	
Max gross soil brg =	0.920 ksf	about N/S axis =	240 K-sec2-ft
Min gross soil brg =	0.850 ksf	about E/W axis =	181 K-sec2-ft
		about Vert axis =	155 K-sec2-ft
N/S eccentricity =	0.003 %		
E/W eccentricity =	0.650 %		

Base Stiffness Results:

@ center of stiffness:	stiffness:	damping ratio:
N/S translation	62686 K/ft	38 %
E/W translation	62686 K/ft	38 %
vertical	81723 K/ft	63 %
about N/S axis	3913509 K-ft/rad	18 %
about E/W axis	3137151 K-ft/rad	17 %
torsional	4309210 K-ft/rad	14 %



SAMPLE DESIGN 1 :
CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN

COMPUTER OUTPUT. SVAP. (cont'd)

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Embedment stiffness results:

@ center of stiffness:	stiffness:	damping ratio:
N/S translation	32698 K/ft	48 %
E/W translation	32698 K/ft	48 %
vertical	17439 K/ft	33 %
about N/S axis	1819361 K-ft/rad	3 %
about E/W axis	1596594 K-ft/rad	8 %
torsional	0 K-ft/rad	0 %

Total Stiffness Results:

Center of stiffness:		Rocking Correction Factors:	
N/S coordinate =	N 0.000 ft	Nr (about N/S) =	1.542
E/W coordinate =	E 0.000 ft	Nr (about E/W) =	1.519
Elevation =	100.000 ft		

@ center of stiffness:	stiffness:	damping ratio:
N/S translation	95384 K/ft	60 %
E/W translation	95384 K/ft	60 %
vertical	99162 K/ft	95 %
about N/S axis	5732871 K-ft/rad	25 %
about E/W axis	4733745 K-ft/rad	29 %
torsional	4389210 K-ft/rad	19 %



SAMPLE DESIGN 1 :

CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN

COMPUTER OUTPUT, SVAP, (cont'd)

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Results of modal analysis:

Mode number:	1	2	3	4	5	6
Natural freq (rpm)	1027	1042	1299	1592	1805	1917
Resonant freq (rpm)	1486	1456	O.D.	1654	2028	2189
Frequency ratio	1.01	1.03	O.D.	0.91	0.74	0.69
Dynamic Magnification	0.53	0.56	0.45	2.65	1.62	1.51
Mode shapes:						
N/S trans (ft)	1.00	0.00	0.00	0.01	0.00	1.00
E/W trans (ft)	0.00	1.00	0.03	0.00	1.00	0.00
Vertical trans (ft)	0.00	-0.03	1.00	0.00	-0.03	0.00
about N/S axis (rad)	0.00	0.09	0.00	0.00	-0.26	0.00
about E/W axis (rad)	-0.09	0.00	0.00	0.03	0.00	0.33
Torsional axis (rad)	0.00	0.00	0.00	1.00	0.00	-0.01

Double Amplitudes at Point 1

Case:	N/S: (mils)	E/W: (mils)	Vert: (mils)
N/S translation	0.28	0.00	0.00
E/W translation	0.00	0.01	0.00
Vert translation	0.00	0.00	0.13
Rocking about N/S axis	0.00	0.01	0.00
Rocking about E/W axis	0.11	0.00	0.00
Torsional	0.00	0.00	0.00

Total @ 1500 rpm	0.39	0.02	0.13
Coastdown @ 1486 rpm	0.39	0.02	0.13
Coastdown @ 1456 rpm	0.39	0.02	0.13



SAMPLE DESIGN 1 :
CENTRIFUGAL COMPRESSOR FOUNDATION STEADY STATE VIBRATION AMPLITUDES & COAST DOWN

COMPUTER OUTPUT. SVAP. (cont'd)

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Double Amplitudes at Point 2

Case:	N/S: (mils)	E/W: (mils)	Vert: (mils)
N/S translation	0.28	0.00	0.00
E/W translation	0.00	0.01	0.00
Vert translation	0.00	0.00	0.13
Rocking about N/S axis	0.00	0.00	0.01
Rocking about E/W axis	0.06	0.00	0.06
Torsional	0.05	0.02	0.00
	-----	-----	-----
Total @ 1500 rpm	0.39	0.03	0.20
Coastdown @ 1486 rpm	0.38	0.03	0.20
Coastdown @ 1456 rpm	0.38	0.03	0.20

End of Data.